

Machine learning-based regional flood frequency framework for climate resilient infrastructure

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ABSTRACT

Traditional design flood estimation approaches assume stationary conditions and rely on historical climate data, potentially under- or over-estimating flood risks. This study proposes a regression-based Regional Flood Frequency Analysis (RFFA) framework to assess projected changes in spring flood quantiles under various warming scenarios. The methodology integrates five distinct regression models using Bayesian Model Averaging (BMA) to establish functional relationships between flood quantiles and catchment physio-climatic variables. Focusing on Ontario's prevalent spring floods, the BMA-RFFA framework demonstrates satisfactory predictive performance across flood quantiles. Future flood quantiles are projected using climatic data from CanRCM4 large ensemble simulations, capturing ensemble spread under a single model and scenario. Results indicate a nearly even distribution of median flood quantile changes from -13% to $+20\%$ under different warming scenarios, with increases in southern and western regions and decreases in the central region, highlighting spatial variability. Climatic variables such as seasonal water storage and rainfall intensity are identified as key predictors of future flood behavior. Continuous design flood changes over 10-year intervals from 1950 to 2100 were derived using an ensemble pooling approach, capturing the temporal evolution of flood quantiles driven by climate ensemble variability. The study also examined the contributions of Anthropogenic Climate Change (ACC) and Internal Climate Variability (ICV), revealing ICV as dominant in most catchments, while ACC shows stronger influence in western regions. Overall, the proposed framework provides enhanced estimation of future flood quantiles and a robust assessment of their changing behavior under projected climate variability, supporting improved flood risk assessment and infrastructure resilience.

1. Introduction

Flood frequency analysis (FFA) is a widely adopted method in hydrologic hazard and risk assessments, particularly for the design of critical infrastructure such as small- to medium-sized dams and levees (Dunne and Leopold, 1978; Hamed and Rao, 2019; Hu et al., 2020; Islam et al., 2024; Kidson and Richards, 2005). Traditional FFA techniques estimate design floods by treating historical extreme events as realizations of a stochastic process that is assumed to be independent, identically distributed (i.i.d.), and stationary (Chow et al., 1988; Hamed and Rao, 2019). However, the validity of the stationarity assumption is increasingly questioned due to the growing evidence of anthropogenic climate change impacts on hydrologic extremes (Fereshtehpour et al., 2025; Koutsoyiannis, 2023; Intergovernmental Panel on Climate Change (IPCC), 2021). Additionally, limited record lengths and observational uncertainties further contribute to the challenges of design flood

estimation, particularly in data-scarce regions (Boni et al., 2007; Gao et al., 2017).

RFFA has long been employed to pool data across hydrologically homogeneous regions and establish functional relationships between physio-climatic characteristics of catchments and flood quantiles, to estimate design floods at desired locations (Griffis and Stedinger, 2007; Ouarda et al., 2008; Hosking and Wallis, 1997). Current approaches range from index-flood method (Burn, 1990; Hailegeorgis and Alfredsen, 2017; Dalrymple, 1960; Hosking and Wallis, 1997) to direct regression approaches (Aziz et al., 2014; Desai and Ouarda, 2021; Gizaw and Gan, 2016; Griffis and Stedinger, 2007) and spatial hierarchical Bayesian modeling (Bracken et al., 2018; Lima et al., 2016; Najafi and Moradkhani, 2013; Sampaio and Costa, 2021; Yan and Moradkhani, 2015). The index flood approach involves the development of a regional curve by averaging all at-site dimensionless frequency curves and establishing a functional relationship generally between the mean

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annual flood to catchment area for determining flood quantiles at both gauged and ungauged sites in the homogeneous region. Hosking and Wallis (1993) employed regional growth curve based on the method of L-moments (Hosking, 1990) that can be directly interpreted as measures of scale and shape of probability distributions. A key advantage of this method is its ability to provide robust flood quantile estimates (e.g., 50-year and 100-year floods) at gauged sites, thereby reducing sampling uncertainty, while also estimating or enhancing flood quantiles at ungauged or poorly gauged sites within a homogeneous region.

The direct regression-based approach estimates at-site flood quantiles for all sites within a homogeneous region and establishes functional relationships between flood quantiles and selected physiographic and climatic variables (Ahn and Palmer, 2016; Haddad and Rahman, 2011; Ouali et al., 2016; Rahman et al., 2020). Its main limitation involves estimating at-site flood quantiles with shorter or limited spans of observed data series at any site. Various regression forms have been employed in previous studies, including ordinary least squares (OLS) (Haddad et al., 2011; Leclerc and Ouarda, 2007; Pandey and Nguyen, 1999), generalized least squares (GLS) (Ahn and Palmer, 2016; Duracher et al., 2016; Haddad et al., 2011; Haddad and Rahman, 2011), support vector regression (SVR) (Allahbakhshian-Farsani et al., 2020; Gizaw and Gan, 2016; Sharifi Garmdareh et al., 2018), random forest regression (RFR) (Desai and Ouarda, 2021), multivariate adaptive regression splines (MARS) (Allahbakhshian-Farsani et al., 2020; Msilini et al., 2022) and Neural Networks (Aziz et al., 2014), among others. While the majority of studies have used single regression techniques, a few comparative analyses have been conducted (Allahbakhshian-Farsani et al., 2020; Haddad et al., 2011; Pandey and Nguyen, 1999). Pandey and Nguyen (1999) found that non-linear models outperformed linear models when the number of predictors was limited, although this improvement often came at the cost of increased predictive uncertainty, a recognized trade-off particularly under limited calibration data (Hastie et al., 2001). Similarly, Haddad et al. (2011) reported that GLS outperformed OLS due to its ability to account for inter-station correlation, enhancing the robustness of regional flood estimations. Allahbakhshian-Farsani et al. (2020) reported that SVR and MARS showed enhanced performance in semi-arid regions compared to other machine learning based models. Multi-modeling can enhance flood prediction by improving model generalization and providing quantifiable measures of prediction uncertainty (Huo et al., 2019; Najafi and Moradkhani, 2014, 2013; Raftery et al., 2005; Schmeits and Kok, 2010; Slougher et al., 2007). However, comprehensive multi-modeling frameworks within the RFFA context, designed to strategically integrate the strengths of diverse regression techniques, remain insufficiently explored, particularly for improving the reliability of future flood quantile estimates.

Rising greenhouse gas emissions are intensifying hydrological processes, leading to elevated flood risks in many regions around the world (Pall et al., 2011; Satterthwaite, 2009; Warden et al., 2024; Stocker et al., 2013). While spatiotemporal dependence can influence hydroclimatic extremes (Koutsoyiannis, 2023; Serinaldi and Kilsby, 2018), attribution studies increasingly suggest that anthropogenic forcings have altered the likelihood of extreme flood events, highlighting the need to revisit traditional stationarity assumptions in design practices. For example, Gillett et al. (2022) found that human-induced climate change played a significant role in devastating floods and landslides in southwestern British Columbia in November 2021. Moreover, studies by Teufel et al. (2017) and Szeto et al. (2015) attributed the severity of the 2013 southern Alberta flood events and the 2014 Prairie flood events to anthropogenic influences. These events underscore the growing influence of climate change on extreme flooding across Canada.

Addressing nonstationarity in hydroclimatic extremes, such as intense precipitation (Kim et al., 2020; Vu and Mishra, 2019; Westra et al., 2014) and streamflow discharge (El Adlouni et al., 2007; Guo et al., 2023; Han et al., 2022; Khaliq et al., 2006) is critical to avoid systematic under- or overestimation of design floods under changing climate conditions. However, deductive approaches to modeling

nonstationarity, such as identifying trends based on theoretical reasoning or externally imposed assumptions, are often practically limited due to data limitations and system complexity. Instead, conditional models that incorporate stochastic covariates must be rigorously validated to distinguish true nonstationarity signals from artifacts of spatiotemporal dependence (Koutsoyiannis, 2023; Serinaldi and Kilsby, 2018). This is particularly relevant when incorporating trends from General Circulation Models (GCMs), Regional Climate Models (RCMs), or hydrological simulations as covariates, where distinguishing externally forced changes from internal variability remains a key challenge.

Ignoring nonstationarity can result in under- or overestimated design floods, leading to infrastructure failure risks or excessive construction costs (Byun and Hamlet, 2020; Cheng and AghaKouchak, 2014; Fereshtehpour and Najafi, 2025; Khaliq et al., 2006). To improve flood estimation in the context of evolving hydrological processes, current approaches involve adopting projections from GCMs or RCMs (Gilroy and McCuen, 2012; Hirabayashi et al., 2013; Li et al., 2019) or employing conditional flood frequency models that incorporate time-varying covariates such as temperature, atmospheric moisture, or climate oscillation indices (AghaKouchak et al., 2012; El Adlouni et al., 2007; Menéndez and Woodworth, 2010; Salas and Obeysekera, 2014). Unlike process-based FFA, which derives future flood levels by driving hydrological models with climate projections (Gilroy and McCuen, 2012; Hirabayashi et al., 2013; Li et al., 2019), RFFA typically links the parameters of statistical distributions to explanatory variables (Leclerc and Ouarda, 2007; O'Brien and Burn, 2014; Villarini et al., 2009). However, attributing trends in flood behavior to changes in covariates requires more than correlation; it demands identification of physical mechanisms driving those trends (Serinaldi et al., 2018). A key limitation remains the scarcity of long-term observational data for robust parameter estimation (Ouarda and Charron, 2019), and projecting future nonstationarity based on historical relationships remains uncertain due to evolving hydrological interactions. Gizaw and Gan (2016) and Awasthi et al. (2022) utilized a direct regression-based flood frequency approach to project changes in flood quantile estimates, incorporating downscaled outputs as predictors from multiple GCMs. Their study highlighted considerable variability in future flood quantile predictions due to the unique structures and formulations of different GCM models. While GCMs provide large-scale climate projections, they often struggle to capture finer regional climate processes (RahimiMovaghar et al., 2025). In contrast, RCMs offer improved resolution (Giorgi, 2019) despite their outputs inherit uncertainties from downscaling assumptions and boundary conditions (Maraun, 2016).

In addition to the effects of Anthropogenic Climate Change (ACC), Internal Climate Variability (ICV), which arises from the inherent chaotic dynamics of the coupled atmosphere–ocean–land system, can substantially influence design flood estimates through its modulation of hydroclimatic extremes across interannual to multidecadal timescales (Feldstein, 2000; Fereshtehpour et al., 2025; Schneider and Kinter, 1994). While ACC drives long-term shifts in the mean state of the climate system and can also alter its variability (Giorgi and Bi, 2009; Lehner et al., 2017; Yin et al., 2018), ICV manifests as natural internal fluctuations that generate regional climate anomalies lasting from months to several decades (Delworth et al., 1993; Kwon and Deser, 2007; Nguyen et al., 2018). Additionally, ACC can influence the variability of hydroclimate variables, albeit with weaker effects compared to ICV (Deser et al., 2012). Thus, when estimating flood levels at the multi-decadal timescale, common in climate impact studies, it is essential to account for both ACC and ICV to assess projected changes and their associated uncertainties (Deser et al., 2012; Kay et al., 2015). The concept of 'Time of Emergence' (TOE) is commonly used to discern when the ACC signal surpasses ICV noise, typically measured using the signal-to-noise (S/N) ratio. This concept aids in assessing climate change impacts and developing adaptation strategies (Faghih and Brissette, 2023; Gaetani et al., 2020; Gu et al., 2019; King et al., 2015). Recent studies have investigated the contribution of ACC and ICV to changes in air temperature and

precipitation separately (Deser et al., 2016; Deser et al., 2012) or in combination (Dai and Bloecker, 2019). However, to the authors' knowledge, no study to date has systematically examined the relative contribution of ACC and ICV to flood levels within the framework of RFFA, despite its importance for a better understanding of climate change impacts (Faghih and Brissette, 2023).

This study addresses the critical need for estimating future flood quantiles at different warming levels, quantifying the corresponding ensemble spread of flood quantiles conditioned on a single model structure and emission scenario, and assessing climate change impacts, especially for designing infrastructure like small- to medium-sized dams and levees. To fill this gap, a regression-based RFFA framework that captures the time-varying behavior of flood quantiles using stochastic climate covariates is developed in three main steps. The first part focuses on developing and validating the RFFA framework by establishing functional relationships between flood quantiles, estimated using Hosking's L-moments, and catchment physio-climatic variables through five distinct regression models. The models are integrated using the Bayesian Model Averaging (BMA) approach, followed by model comparison and validation. The second part quantifies changes in future flood quantiles under three different warming levels (1.5 °C, 2 °C, and 3 °C) using CanRCM4-LE simulations. While ensemble members support probabilistic analyses of internal variability, they do not constitute a fully comprehensive uncertainty framework, as they are conditioned on a single climate model and emission scenario and do not account for structural or scenario uncertainty (Deser, 2020; Koutsoyiannis and Montanari, 2022). This section also evaluates continuous changes in design floods at 10-year intervals from 1950 to 2100, addressing the temporal evolution of flood quantiles driven by climate ensemble variability under climate change. The final part examines the relative roles of ACC and ICV in future flood quantile estimation. To present the findings coherently, Section 2 details the methodology, Section 3 discusses the results and discussion, and Section 4 provides the conclusions of the study.

2. Dataset and study area

This study develops a regression-based multi-modeling framework for simulating spring flood quantiles across multiple catchments in southern Ontario under climate change impacts. The focus on spring floods is due to the predominance of recorded flood events during this season (Javelle et al., 2003; Singh J. et al., 2021). The model development relies on two types of data: the physiographic characteristics of the catchments and time-varying climatic data. The physiographic characteristics are extracted from a digital elevation model with 30 m spatial resolution utilizing Ontario Watershed Information Tool (Ontario watershed information tool (OWIT), 2022). Climatic data for model development are obtained from ERA5 reanalysis, while future changes and temporal evolution of flood quantiles are assessed using projections from the CanRCM4 Large Ensemble (LE).

2.1. ERA5 reanalysis data

The ERA5 reanalysis data provides records of the global atmosphere, land surface, and ocean waves at single level and different vertical levels dating back to 1940 until the present. It is the latest global reanalysis dataset provided by the European Center for Medium-Range Weather Forecasts (ECMWF) (Hersbach et al., 2020). This dataset has horizontal resolution of 0.25 degrees and 1-h frequency. For this study, single level climate data have been extracted by averaging the corresponding values from grids that cover the catchment area and hourly data converted to daily scale. In cases where catchment dimensions are smaller than a single grid or when a single grid is covered by multiple catchments, grid information corresponding to the majority of the catchment area is selected for that particular catchment. Initially six climatic features are extracted for each basin based on the daily ERA5 single level data from

1986 to 2016. These features include seasonal total rainfall, seasonal total number of days exceeding 10-mm rainfall (R10), T-year daily rainfall intensity, seasonal soil moisture, seasonal snow water equivalent (SWE) and seasonal potential evapotranspiration (PET). The spring flood season (March-May) is considered for rainfall, soil moisture, and PET, while SWE is defined over the winter season (December-February) to capture snow accumulation and melt.

2.2. CanRCM4-LE

This study employs CanRCM4-LE simulations to project flood levels. CanRCM4-LE is a regional Single Model Initial-condition Large Ensemble (SMILE) developed by the Canadian Centre for Climate Modeling and Analysis (CCCMA) of Environment and Climate Change Canada. It covers North America at a spatial resolution of 0.44° and a daily temporal scale for all ensemble members and sub-daily scale for some selected ensemble identifiers (Scinocca et al., 2016). For both the historical (1950–2005) and future (2006–2100) periods under the RCP8.5 scenario, 50-member large ensemble simulations are generated, representing a business-as-usual and non-climate policy scenario. Very small perturbations, with the same external forcing and model physics, are applied to each member of CanRCM4-LE. Consequently, all members share the same radiative forcings, and differences between realizations are attributed to internal climate variability. A pooling approach is employed, combining data from all 50 members of CanRCM4-LE. This enables robust statistical modeling with a significantly increased sample size (Ben Alaya et al., 2020; Innocenti et al., 2019; Singh H. et al., 2021). The extraction of climate variables from CanRCM4-LE follows the same methodology as employed for ERA5.

2.3. Study area

The study area chosen to evaluate the performance of our proposed RFFA model and to project future flood changes comprises a group of river catchments in Ontario, Canada. This selection is motivated by the region's historical vulnerability to flood damage, which has seen an increasing frequency of recorded floods on a national scale (Zahmatkesh et al., 2019). Furthermore, Ontario is home to approximately 40 % of Canada's population and contains one of the highest concentrations of dams in the country, ranking just behind Quebec in terms of total number of dams (CDA, 2013), with many located around the Great Lakes. Since flood frequency analysis is recommended for catchments with natural conditions or minimal regulation, as outlined in USGS Bulletin 17C (England et al., 2019), our approach incorporates surrounding catchments that are free from major dams or significant flow modifications. The basin selection process involves screening daily discharge data records over a minimum of 20-year period between 1981 and 2020, focusing on catchments with drainage areas ranging from 50 to 2500 km² (Fig. 1(c)). This study focuses on spring floods, as the majority of flood events in the region occur during this season (Fig. 1(b)) and are influenced by season-specific climatic variables. Each year is defined by the spring season (March-May) to extract seasonal maximum discharge values, and a year is included if more than 90 % of daily data are available during this period. To ensure temporal consistency in the streamflow records, we applied both the Mann-Kendall trend test (Mann, 1945) and the Pettitt change point test (Buishand, 1982) to the spring flood series for each station. Stations exhibiting significant trends ($p < 0.05$) or abrupt shifts in the discharge records were excluded from the analysis. Out of the initially selected stations, 8 showed significant monotonic trends and 6 exhibited statistically significant change points. To establish a homogeneous region, a data discordancy test with heterogeneity measures recommended by Hosking and Wallis (1997) is applied, resulting in the final selection of 114 catchments for this study (Fig. 1(a)). This region is characterized by relatively flat terrain and a humid continental climate. Annual average rainfall ranges between 690 mm to 1220 mm for the selected stations.

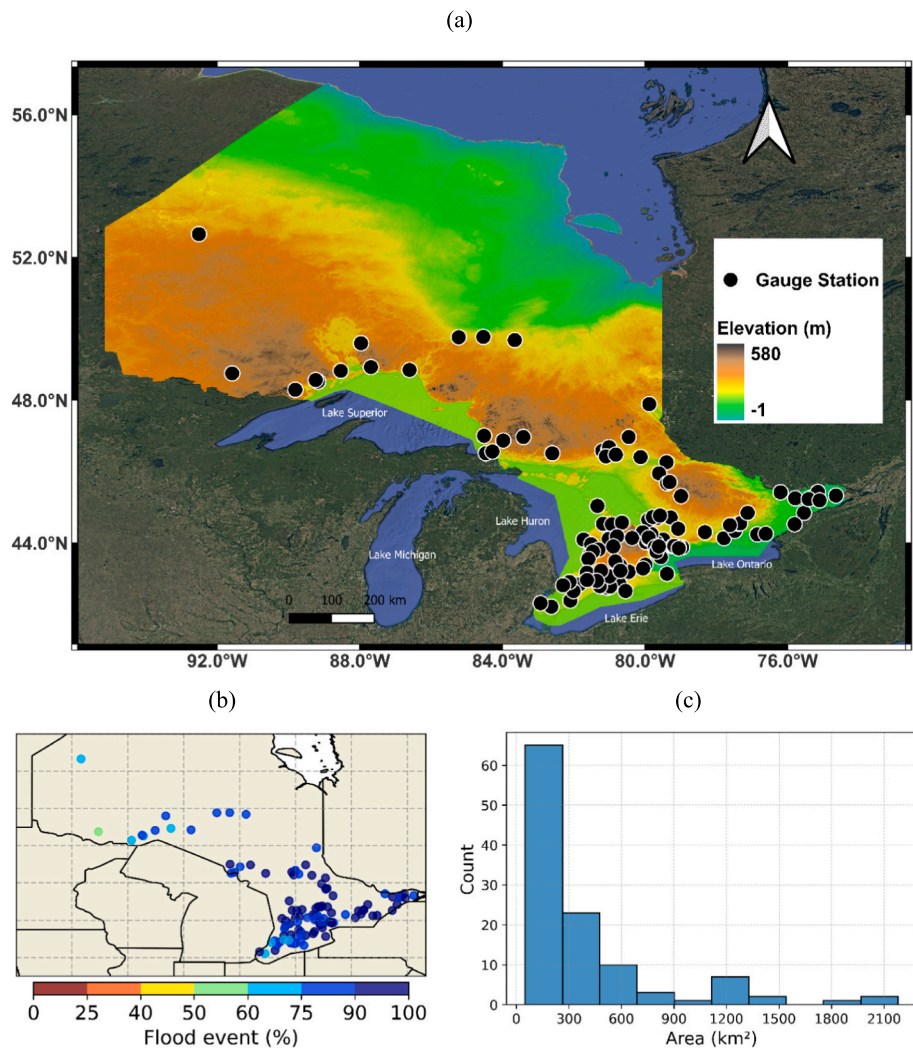


Fig. 1. Geographic location of the study area (a) Spatial distribution of 114 hydrometric gauge stations across Ontario, Canada overlaid on a 30-arc second DEM background. (b) Percentage of total flood events occurring during the spring season at each station. (c) Histogram showing the distribution of gauge stations categorized by drainage area.

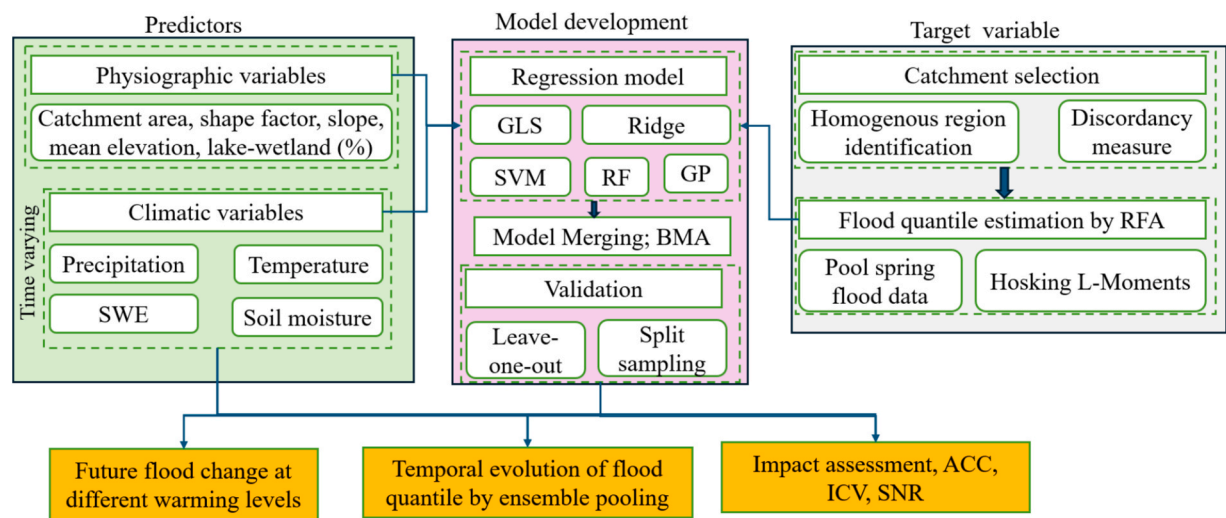


Fig. 2. Methodological framework for the regression-based BMA-RFFA model. The model is utilized to estimate future flood quantile changes under various warming levels, address temporal evolution of flood quantiles using ensemble pooling, and assess the relative contributions of ACC and ICV through signal-to-noise ratio (SNR) analysis across different warming scenarios.

3. Methodology

This study develops a multi-model design flood estimation methodology by integrating the Hosking L-moments-based RFFA approach to extract various flood quantiles (10-year, 25-year, 50-year, and 100-year) for the selected catchments and establish functional relationships with catchment physiographic and climatic variables using the regression-on-quantiles technique (Fig. 2). The Hosking L-moments-based RFFA approach has been preferred over at-site flood quantile estimation to reduce sampling uncertainty, particularly for longer return periods in regions with limited observational records or significant data gaps. Five distinct regression models are utilized, including generalized least squares (GLS), Ridge regression, support vector machine (SVM), random forest (RF), and Gaussian process (GP) regression. Each machine learning-based regression model is optimized using a grid-search approach for hyperparameter tuning. The selection of regression models is based on their widely recognized performance in the hydro-climatic domain, particularly in RFFA, and the unique strengths of their underlying algorithms. These regression modelling approaches are integrated by using Bayesian model averaging (BMA) and evaluated through leave-one-out cross validation (LOOCV) and split sampling approach to assess their performance for ungauged stations and unseen time horizons, respectively. Furthermore, this study investigates future flood changes and their impacts under different warming scenarios. Flood-relevant projections of climatic fields, derived from a large ensemble of regional climate model simulations (Scinocca et al., 2016), are integrated with the multimodel approach to develop future projections of design floods. In the subsequent section, the relationship between estimated flood quantiles and catchment characteristics, along with time-varying climatic variables, is established.

3.1. Predictor selection

To construct the functional relationship between flood quantiles and physio-climatic variables of catchments through regression modelling for future flood projections, a set of physiographic and climatic predictors is selected based on insights from previous studies (Durocher et al., 2015; Gizaw and Gan, 2016; Haddad and Rahman, 2012; Han et al., 2020; Merz and Blöschl, 2005) supplemented by input from engineering expertise. In Ontario, spring flooding typically results from the combined effects of snowmelt, antecedent soil moisture, and extreme rainfall. Therefore, two crucial factors for spring flood generation: SWE and seasonal soil moisture are included (Berghuijs et al., 2019; Buttle et al., 2016; Musselman et al., 2018), as they are often overlooked in previous RFFA studies. Other climatic variables include seasonal total rainfall, seasonal R10, rainfall intensity for the T -Year return period, and seasonal PET. The physiographic variables encompass the basin's centroid latitude, centroid longitude, drainage area, shape factor, length, slope, mean elevation, time of concentration and the percentage of lake wetland coverage. To address potential multicollinearity issues, where high intercorrelation can inflate standard errors and thereby reduce the likelihood of detecting statistically significant relationships, a correlation matrix analysis was first conducted followed by Least Absolute Shrinkage and Selection Operator (LASSO) regression with cross-validation (Bardsley et al., 2015; Freijeiro-González et al., 2022; Tibshirani, 1996). LASSO improves model stability by penalizing the absolute size of regression coefficients, effectively performing variable selection by shrinking some coefficients to zero when predictors are redundant or highly correlated (Kutner et al., 2005).

The predictor importance across different regression models for estimating flood quantiles is evaluated using SHapley Additive exPlanations (SHAP) method. The SHAP method is grounded on Shapley values (Shapely, 1953), a concept from cooperative game theory that assigns a value to each feature or variable in a prediction model based on its contribution to the model's prediction. SHAP has been used to interpret the output of machine learning models in various applications,

including hydrology (Dikshit and Pradhan, 2021; Sushanth et al., 2023), particularly on RFFA study (Mangukiya and Sharma, 2024). The SHAP values of the features are analyzed to determine their importance in predicting flood quantiles.

3.2. Homogeneous region identification, goodness-of-fit test, and regional frequency analysis

The RFFA is guided by the identification of hydrologically homogeneous regions and the selection of regional frequency distribution. A homogeneous region is established using the discordancy measure recommended by Hosking and Wallis (1997). In the screening test, discordant hydrometric stations are identified from the total hydrometric stations based on the difference of the L -moment ratios of a hydrometric station from the regional average L -moment ratios of all the hydrometric stations (Hosking and Wallis, 1997). A site is considered discordant if the discordancy measure, denoted as D_i , is greater than or equal to 3.0 (Hosking and Wallis, 1997).

$$D_i = \frac{1}{3} [N(u_i - \bar{u})^T S^{-1} (u_i - \bar{u})], i = 1, 2, 3 \dots N \quad (1)$$

where $S = \sum_{i=1}^N (u_i - \bar{u})(u_i - \bar{u})^T$ and $\bar{u} = \frac{\sum_{i=1}^N u_i}{N}$. u_i contains the estimates of sample L -moment ratios of site i , $\bar{u}$ is the unweighted group average, and N is the number of sites in the region.

To identify homogeneous region, Hosking and Wallis (1997) introduced heterogeneity measures (statistics) based on the theory of L -moments. These measures compare the regional dispersion of L -moment ratios with the average dispersion of the L -moment ratios determined from simulations of homogeneous groups from a four-parameter Kappa distribution influenced only by sampling variability. The statistic to assess heterogeneity in a group of sites is given by:

$$H = \frac{(V - \mu_v)}{\sigma_v} \quad (2)$$

where $V = \left[\frac{\sum_{i=1}^N n_i (t^{(i)} - t^R)^2}{\sum_{i=1}^N n_i} \right]^{\frac{1}{2}}$. Here, μ_v is the mean and σ_v is the standard deviation of the computed inter-site variation obtained through simulations. n represents the number of years, t is the sample L -CV, $t^{(i)}$ is L -moment ratio, and $t^R = \left[\frac{\sum_{i=1}^N n_i t^{(i)}}{\sum_{i=1}^N n_i} \right]$ represents the regional average L -CV.

According to Hosking and Wallis (1997), a region can be considered "acceptably homogeneous" if $H < 1$.

The selection of frequency distributions is guided by goodness-of-fit measures, which assess how well the candidate distributions represent the observed data. For the regional frequency analysis, we tested four widely used three-parameter distributions: Generalized Extreme Value (GEV), Generalized Logistic (GLO), Generalized Normal (GN), and Pearson Type III (PE3), each defined by location, scale, and shape parameters. These distributions were selected due to their frequent application in flood frequency analysis (for example, Hosking and Wallis, 1997; Zhang et al., 2020), their flexibility in representing skewed and heavy tailed behavior, and their adoption in various national and regional flood estimation guidelines. The common goodness-of-fit criterion for various candidate distributions in terms of the L -moments is based on the Z -statistics. Computation of the Z -statistics involves fitting of a four-parameter Kappa distribution to the regional average L -moment ratios t_1^R , t_2^R , t_3^R and t_4^R and then simulation of a large number of realizations of a region with N sites. For each candidate distribution, the goodness-of-fit measure is calculated from the regional average L -kurtosis $t_4^{[m]}$, the bias and standard deviation of t_4^R for the m^{th} simulated region (Hosking and Wallis, 1997). The distribution (DIST) is declared to fit adequately if Z_{DIST} is sufficiently close to zero, a reasonable criterion being $|Z_{DIST}| \leq 1.64$ (Hosking and Wallis, 1997).

Hosking and Wallis (1993, 1997) considered an index flood procedure in which the parameters are estimated separately at each site. They recommend the use of a weighted average of the at-site estimates:

$$\hat{\theta}_k^R = \frac{\sum_{i=1}^N n_i \hat{\theta}_k^i}{\sum_{i=1}^N n_i} \quad (3)$$

where N is the number of stations, $\hat{\theta}_k^i$ is the L -moment of interest, and n_i is the number of data values at each station. Substitution of these estimates into $\hat{q}(F)$ gives the estimated regional quantile $\hat{q}(F) = \hat{q}(F; \hat{\theta}_1^R, \dots, \hat{\theta}_p^R)$ (Hosking and Wallis, 1993). The at-site i quantile estimates are obtained by combining the estimates $l_1^{(i)}$ and $\hat{q}(F)$:

$$\hat{Q}_i(F) = l_1^{(i)} \hat{q}(F) \quad (4)$$

where $\hat{Q}_i(F)$ are at-site flood quantiles for site i , $l_1^{(i)}$ is average of annual maximum flood at site i , and $\hat{q}(F)$ are regional flood quantiles for different return periods. For more details on the index flood method, readers are referred to Dalrymple (1960). Further information on L -moments can be found in Hosking and Wallis (1993, 1997).

3.3. RFFA models

The most commonly used relationship between the flow quantile and the basin characteristics is described by the power-form function (Pandey and Nguyen, 1999; Thomas and Benson, 1970).

$$Q_T = \alpha_0 A_1^{\alpha_1} A_2^{\alpha_2} \dots A_p^{\alpha_p} \varepsilon_0 \quad (5)$$

where Q_T represents the flood quantile, $(A_1, A_2, \dots, A_p)$ are basin physiographic and climatic characteristics, $(\alpha_1, \alpha_2, \dots, \alpha_p)$ are the model parameters, ε_0 is the multiplicative error term, and p is the number of basin characteristics considered for the study. The power-form model can be linearized by a logarithmic transformation.

$$\log(Q_T) = \log(\alpha_0) + \alpha_1 \log(A_1) + \alpha_2 \log(A_2) + \dots + \alpha_p \log(A_p) + \log(\varepsilon_0) \quad (6)$$

Eq. (6) establishes the nonlinear relationship between basin characteristics and flood quantile in a linear format, forming the fundamental basis for the regression model. In this study, five regression models are independently applied and their outputs are integrated using BMA. Generalized Least Squares (GLS) regression, widely used in RFFA for handling cross-correlated flows and variable record lengths (Haddad and Rahman, 2011). Ridge regression, which addresses multicollinearity through coefficient regularization (Hoerl and Kennard, 1970). To capture complex non-linear relationships between flood quantiles and climate/physiographic variables, Random Forest (RF), Support Vector Machine (SVM) regression, and Gaussian Process (GP) regression are incorporated. These models offer enhanced flexibility, with features such as built-in variable importance evaluation features in RF and predictive uncertainty quantification in GP regression model. A concise overview of each model is described in the Supplemental Section (See Text S1).

3.4. Integration of multiple RFFA models

In contrast to the commonly employed “single model approach”, integrating multiple models for RFFA enhances predictive performance, and improves generalization across diverse catchments. An additional advantage of model integration lies in its ability to quantify inter-model uncertainty; however, this aspect is not the primary focus of the current study. Individual regression models may capture specific aspects of the data while potentially overlooking intricate relationships that influence outcomes. By integrating diverse model strengths and compensating for

their weaknesses, we achieve a more comprehensive representation of the data's underlying structure. In our study, we utilize Bayesian Model Averaging (BMA) to merge five regression models. This method addresses uncertainty in the total variance through the predictive probability distribution function. The BMA predictive PDF for a quantity of interest, in this study the flood quantile, is a weighted average of the individual model PDFs (Duan et al., 2007; Najafi et al., 2011). The BMA scheme is derived from the law of total probability and can be expressed as follows:

$$p(y) = \sum_{i=1}^n p(m_i|O) \bullet p(y|m_i, O) \quad (7)$$

where O represents the observed data, y is predictand, and m_i is the prediction from model i . $p(m_i|O)$ is the posterior probability of model prediction m_i , also referred to as the likelihood of model prediction m_i being the correct prediction given the observed data O . This term indicates how well a specific ensemble member aligns with the observations. It also corresponds to the weights assigned to each model, ensuring that the sum of the weights for all models equals to 1. $p(y|m_i, O)$ is the posterior distribution of y given O and m_i , and it can be approximated by a Gaussian distribution $g(y|m_i, \sigma_i^2)$, where m_i is the mean and σ_i is the standard deviation. The choice of the posterior distribution can be rationalized through the logarithmic transformation applied to the regression model equation and the stochastic nature of the model's residuals. These can be obtained using an optimization procedure such as the Expectation Maximization algorithm (Raftery et al., 2005).

3.5. Model evaluation

To assess the efficacy of the developed flood quantile estimation models, two widely used validation procedures are employed: Leave-One-Out Cross-Validation (LOOCV) and split sampling. The main difference between LOOCV and split sampling lies in their focus: LOOCV assesses model performance for ungauged stations, while split sampling evaluates model performance for unobserved climatic periods. These complementary approaches ensure the robustness of the developed flood quantile estimation models across spatial and temporal contexts. LOOCV method has been widely applied in RFFA studies (Ahn and Palmer, 2016; Han et al., 2022; Ouali et al., 2016; Rahman et al., 2020). This jack-knifing approach initially estimates flood quantiles at a given gauge station using regional information, without incorporating local flood data. In the second step, these regional estimates are compared to the at-site flood quantile (Merz and Blöschl, 2005) derived from both index flood approach under Hosking's L -moment framework and empirical estimates. This iterative process is carried out for each station, and the evaluation metric is calculated upon completion of the process for all stations. Similarly, for applying the BMA to integrate different regression-based RFFA models, we first determine the weights for each model, $p(m_i|O)$, during the training phase. This is done while one station is excluded through LOOCV. Subsequently, in the prediction phase, these computed weights are used to estimate flood quantiles for the target basin.

In the split-sampling approach, the full study period is divided into two distinct intervals: 1980–2002 and 2003–2019. For each interval, at-site flood quantiles for all stations are independently estimated using Hosking's L -moment framework, and time-varying climatic predictors are extracted separately for both periods. Regression models are developed using data from the initial period (1980–2002) and subsequently applied to predict flood quantiles for the later period (2003–2019). The predicted flood quantiles are then evaluated against the at-site flood quantiles derived from Hosking's L -moment framework for the 2003–2019 interval.

For both validation processes, model performance is evaluated using several statistical metrics, including Spearman's rank correlation coefficient (ρ), Mean Absolute Error (MAE), Kling-Gupta efficiency (KGE)

(Gupta et al., 2009), Nash-Sutcliffe coefficient (NSE) (Nash and Sutcliffe, 1970), coefficient of determination (R^2), root mean squared error (RMSE) and mean bias (BIAS). Although KGE and NSE are traditionally used for time-series evaluation, we include them here to enable comparison with prior RFFA studies where such metrics have been employed for evaluating quantile-based estimates. Given the range of watershed drainage areas in this study, from 50 to 2500 km², metrics such as BIAS and RMSE could be overly influenced by errors in flood discharges predicted for large watersheds as compared to small ones. To address this, in addition to BIAS and RMSE, we also apply mean absolute percentage error (MAPE) and relative root mean squared error (RMSEr) (Desai and Ouara, 2021; Gizaw and Gan, 2016).

3.6. Future flood projection and assessment of temporal evolution

After initially training and validating the BMA-RFFA model using historical ERA5 reanalysis data, the model's predictive capacity is extended to project future changes in flood quantiles. This involves an additional training phase using the 50-member CanRCM4-LE climate dataset. The BMA-RFFA model is calibrated using the baseline period (1986–2016), utilize the same set of hyperparameters optimized during the machine learning model validation stage and subsequently applied to project future flood quantiles for each ensemble member of CanRCM4-LE under three global warming scenarios: +1.5 °C (2024–2054), +2 °C (2034–2064), and +3 °C (2053–2083). These projections are based on CanRCM4-LE simulations corresponding to the specified global warming levels, which are derived from CanESM2-LE under historical and high-emission RCP8.5 scenarios (Cannon et al., 2020).

To address the temporal evolution of flood quantiles, a well-established ensemble-pooling approach (Innocenti et al., 2019; Singh H. et al., 2021) is employed for each 10-year block from 1951 to 2100, combining data from all 50 ensemble members (Fig. 3). This method preserves information from all simulations, providing robust flood quantile estimates by aggregating 500 (10 years * 50 members) hydrologic years in each 10-year block. It also ensures negligible trends within shorter 10-year time block, satisfying the statistical assumptions of stationarity and identical distribution within the block (Singh H. et al., 2021). This technique illustrates how flood quantiles evolve every 10 years, highlighting their continuous change over the entire time period. Furthermore, this approach provides a realistic representation of climate variability, as it accounts for different initial conditions used to generate ensemble members.

Unlike traditional methods that focus on specific historical and future periods, this approach captures flood quantile changes continuously, mitigating sampling uncertainty by leveraging large ensemble simulations. The 50-member CanRCM4-LE dataset is used to extract all climatic predictors from the RFFA model for each 10-year block. The BMA-RFFA model is trained using climatic data from each ensemble member for 1986–2016 (baseline period) to prevent bias towards any particular ensemble member and then applied to estimate flood

quantiles for subsequent periods, providing a detailed assessment of evolving flood risks. Thus, 50-member simulated flood quantiles for each 10-year block illustrate ensemble spread, conditioned on a single model structure and emission scenario, representing the temporal evolution of flood quantiles throughout the time period from 1950 to 2100.

3.7. Assessing the relative contributions of ACC and ICV to future flood projections

The role of internal climate variability (ICV) in relation to anthropogenic climate change (ACC) is quantified using the signal-to-noise ratio (SNR) as a diagnostic (Hawkins and Sutton, 2009):

$$SNR = \frac{ACC}{ICV} \times 100\% \quad (8)$$

ACC is defined as the change in the multi-model or multi-members ensemble mean (Nguyen et al., 2018; Sui et al., 2014). To compute ACC, we initially determined the change in different flood quantiles in the future (2071–2100) relative to the base period (1976–2005) for each of the 50 members of CanRCM4-LE. These changes are regarded as flood change signals for each member. Finally, the signal is defined as the arithmetic average across all members (Sui et al., 2014; Nguyen et al., 2018).

$$ACC = \frac{1}{m} \sum_{i=1}^m M_i \quad (9)$$

where m is the number of CanRCM4 members (i.e., 50), and M_i represents the future change in T -year design flood magnitude for the i^{th} member. In this study, we calculate internal-model variance to account for differences across multiple realizations of the same climate model (Deser et al., 2012). ICV, which spans various temporal and spatial scales across different climate variables, is defined here as the variability in design flood magnitudes within a single 30-year window at each station. To assess changes in flood magnitude, we compute absolute differences in T -year flood quantiles between the future period (2071–2100) and the baseline period (1981–2010) for each ensemble member across all gauge stations. The standard deviation of these changes in flood quantiles, calculated from the ensemble members of CanRCM4, serves as a measure of ICV, capturing the spread of internal variability relative to the ensemble mean.

$$ICV = \sqrt{\frac{1}{m} \sum_{i=1}^m (M_i - \bar{M})^2} \quad (10)$$

where $\bar{M}$ is the ensemble mean change.

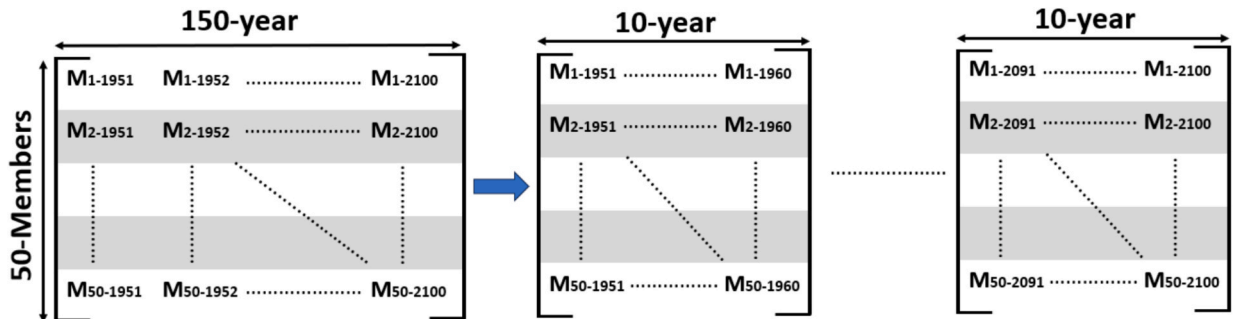


Fig. 3. Data pooling approach at 10-year intervals to represent temporal evolution of flood quantiles driven by climate ensemble variability for 150 years (1951–2100).

4. Results and discussion

4.1. Predictor selection

The initial selection of predictors is guided by the spring flood generation processes and findings from relevant studies as discussed in the previous section. The pair-wise correlation among these selected variables is shown in Fig. S1. The correlation matrix reveals significant cross-correlations among several variables, particularly between river stream length and catchment area, time of concentration and river stream length, time of concentration and catchment area, seasonal SWE and centroid latitude, seasonal SWE and seasonal PET, as well as seasonal PET and seasonal rainfall, all of which exhibited correlation coefficients exceeding ± 0.8 . High correlation between river length and time of concentration is expected, as length is directly used to compute time of concentration (de Almeida et al., 2014), similar to the case between river length and basin area. To address multicollinearity among climatic variables and better represent springtime water availability, a composite variable, seasonal water storage, is introduced by integrating key hydrometeorological components: spring rainfall, spring PET, and winter SWE (Eq. (11)):

$$\text{seasonal water storage} = \text{spring rainfall} + \text{winter SWE} - \text{spring PET} \quad (11)$$

Furthermore, to formally address multicollinearity among the remaining predictors and identify a parsimonious model, we applied LASSO regression with five-fold cross-validation. This regularization technique penalizes model complexity by shrinking non-informative coefficients to zero. The final set of retained predictors (Fig. S2) was selected based on their contribution to minimizing cross-validated error under the penalized regression framework. Additionally, T-year rainfall intensity, which represents rainfall intensity for a given return period, is incorporated to align flood quantile estimates with corresponding return period rainfall events, following previous research (Gizaw and Gan, 2016; Rahman et al., 2020). For example, the model for estimating the 10-year flood quantile is built using the corresponding 10-year rainfall intensity. The finalized predictor set shown in Fig. 4 demonstrates acceptable pairwise correlation (less than ± 0.6). To further validate the importance of each predictor, we conducted recursive feature elimination with cross-validation (RFECV) using a linear regression model (Hastie et al., 2001). This analysis confirmed that each of the nine selected predictors progressively improved cross-validated R^2 , reinforcing their collective value in enhancing model performance (Fig. S3).

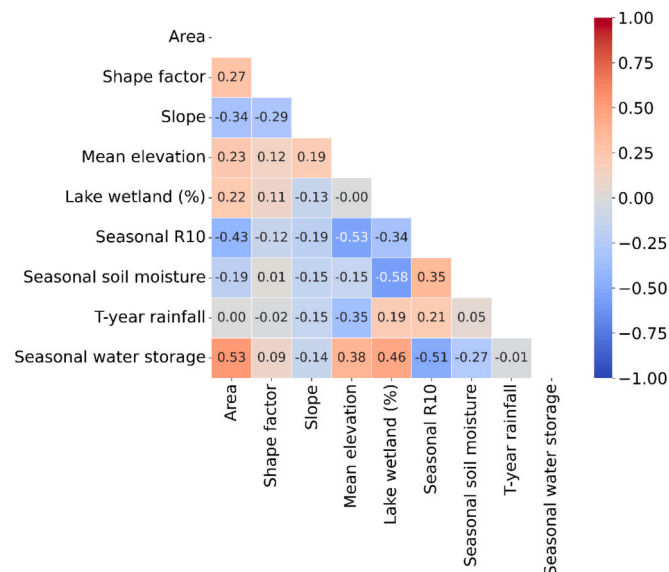


Fig. 4. Correlation matrix between selected covariates.

The final selection of variables and their respective statistical properties are presented in Table 1.

4.2. Homogenous region identification and flood quantile estimation

In the initial stage, 141 catchments in Ontario are screened based on the availability of minimum 20-year annual maximum discharge data from 1980, meeting the drainage area threshold and natural flow conditions. The discordancy measure (D_i) is then calculated, and stations with a D_i value exceeding 3.0 are excluded. This iterative process continues until no station within the group exhibited D_i value above 3.0. The final selection comprises 114 gauge stations, each having a D_i value less than 3.0, with the corresponding L -moments and L -moment ratios presented in Table S1. The heterogeneity measures (H -values) to identify homogenous region and Z^{DIST} statistics were employed to assess the goodness-of-fit of candidate distributions for regional frequency analysis (Hosking and Wallis, 1997) as shown in Table 2. The region is found acceptably homogeneous with an H_1 value of -1.32 , following Hosking and Wallis (1997) recommendation of being below 1.0.

Based on Hosking and Wallis (1997) recommendation for the identification of regional distribution, the GEV distribution has a Z -value (-1.80) closest to the reference threshold ($|Z| < 1.64$), followed by GN, PE3, and GLO distributions. Although the GEV's Z -value marginally exceeded the recommended threshold, it was prioritized for final selection based on its minimal distance in the L -moment ratio diagram (Fig. S4), where it exhibited the closest fit to the regional data's L -skewness and L -kurtosis relationships (Peel et al., 2001; Reinders and Munoz, 2024). Consequently, the GEV distribution is selected for the regional frequency analysis. Furthermore, the selection of the GEV distribution is supported by Zhang et al. (2020), whose Canada-wide analysis demonstrated that it outperformed other commonly used distributions by providing broader applicability and closer alignment with empirical flood quantiles.

Using the L -moments approach outlined by Hosking and Wallis (1993, 1997) within the index flood framework and adopting the GEV distribution, 10-year, 25-year, 50-year, and 100-year flood quantiles are estimated for each gauge station. Concurrently, at-site flood quantiles are calculated using the same distribution. The resulting estimates,

Table 1

List of selected physiographic and climatic variables and their statistical properties.

No.	Type	Variable	Unit	Min.	Median	Max
1	Physiographic	Catchment Area	km ²	50.26	211.88	2178
3		Shape factor	—	3.60	9.26	27.06
4		Slope of mainstream	m/km	0.74	2.39	12.19
5		Mean elevation	m	75.58	282.39	515.85
6		Lake wetland fraction	%	0.16	11.33	64.45
7	Climatic	Seasonal water storage	mm	138.70	210.54	480.51
9		Seasonal R10	days	3.57	7.4	9.4
10		Seasonal soil moisture	mm	115.98	334.7	435.15
11		10-year rainfall intensity	mm/day	28.42	35.17	40.61
12		25-year rainfall intensity	mm/day	31.76	39.54	49.83
13		50-year rainfall intensity	mm/day	33.97	42.99	58.92
14		100-year rainfall intensity	mm/day	35.98	45.78	69.49

Table 2
Summary of homogeneity test statistics and selection of regional distribution through Z^{DIST} goodness-of-fit test.

No of catchment with D(i) < 3.0	Homogeneity test			Distribution selection: Z-value			
	H ₁	H ₂	H ₃	Generalized Logistic (GLO)	Generalized Extreme Value (GEV)	Generalized Normal (GN)	Pearson Type III (PE3)
114	−1.32	−2.47	−1.89	4.72	−1.80	−2.67	−3.87

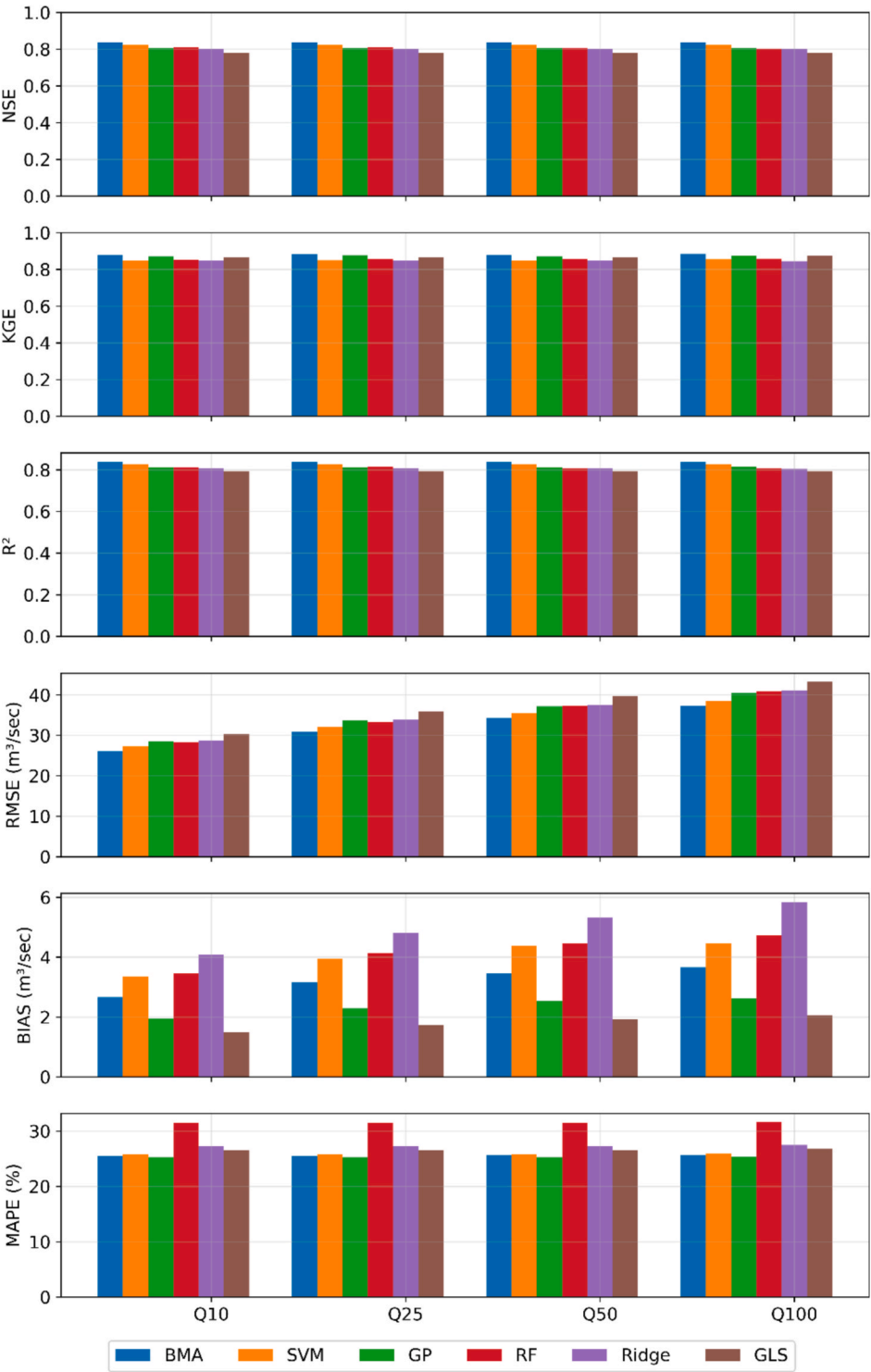


Fig. 5. Comparative evaluation metrics using LOOCV approach for all selected models including five individual RFFA and one integrated BMA-RFFA model with different flood recurrence intervals (10-year, 25-year, 50-year and 100-year). Climate predictors extracted from ERA5 data are used for model development.

presented in Table S2, show that the differences between the regional and at-site methods are minimal for lower return periods (e.g., 10-year) but increase for higher return periods, highlighting the greater sampling uncertainty inherent in at-site methods when dealing with limited data. Consequently, the L-moments-based index flood framework, known for reducing sampling uncertainty (Mosaffaie, 2015) is employed to derive the response variables (10-year, 25-year, 50-year, and 100-year flood quantiles at each station) for regression model development.

4.3. Model development, comparison, integration and evaluation

Five individual regression models and one integrated model using the BMA approach are developed based on the predictor set and response variable estimated in sections 4.1 and 4.2. Predictors are normalized during model development to avoid unit differences and ensure numerical stability in the regression models. Climatic variables extracted from ERA5 data serve as time varying predictors for model development and evaluation stage. Model performance is assessed using LOOCV and split sampling approaches. Fig. 5, and Table S3 present the LOOCV results comparing predicted flood quantiles from each model, while Figs. 6 and S5 specifically show BMA-RFFA predicted flood quantiles against at-site flood quantiles derived from the index flood approach using Hosking's L-moment method and empirical estimates respectively. The evaluation metrics in Figs. 6 and S5 showed consistent agreement for 10-year and 25-year flood quantile regardless the source of at-site reference value. However, agreement was weaker for the 50-year flood when BMA-RFFA predictions were compared with empirical quantiles, likely due to limited data availability, only 42 out of 114 stations have records longer than 50 years, and no station exceeds 100 years (Fig. S5). Generally, model performance declines for basins with larger catchment areas. However, only 26 out of 114 catchments exceed 500 km² in area. As regression model performance depends on available training data, a decrease in accuracy is observed for catchments of extreme sizes (very large) due to limited training data. This limitation of regression-based RFFA models has been acknowledged in previous

studies (Gizaw and Gan, 2016; Ouarda et al., 2008).

Fig. 5 compares the performance of individual and integrated models using multiple evaluation metrics. All selected models demonstrate satisfactory performance, with NSE and R^2 values surpassing 0.78 for different flood quantiles. The BMA model, which integrates all five individual models, outperforms the others across most evaluation metrics. The BMA model assigned the highest posterior weight to GP (0.36), followed by RF (0.29), SVM (0.13), Ridge (0.12), and GLS regression (0.10), as shown in Fig. S6. This distribution reflects the relative contribution of each model based on its likelihood and residual variance structure. Furthermore, the range of inter-model uncertainty (95 % band) of the 5 regression models for 100-year flood prediction shown in Fig. S7. Among evaluation metrics, RMSE values follow a similar pattern to NSE and R^2 , with BMA showing the best performance, followed by SVM, GP, and RF. However, GLS has the lowest overall bias, while BMA and GP show superior performance based on MAPE, which considers absolute deviations irrespective of direction. The BMA model combines predictions from competing models to provide probabilistic predictions. Previous studies have also demonstrated its higher reliability and skill in predicting floods compared to individual models (Duan et al., 2007; Najafi et al., 2011). It consistently performs well in predicting various flood quantiles, with KGE value greater than 0.87 and Spearman's rank correlation coefficient (ρ) values equal to 0.92 (Fig. 6). A similar level of agreement is found in the split sampling evaluation, where KGE values for the BMA model range from 0.79 to 0.84 and ρ remains greater than or equal to 0.94 across different flood quantiles (Fig. S8). The results from the split-sampling approach suggest that the integrated BMA model provides reasonable estimates of various flood quantiles for the studied catchments across different time periods, capturing the variability in climatic predictors.

As shown in Fig. 5, and the scatter plot in Figs. 6, S5, and S8, the BMA model generally performs comparably or better than individual models in estimating T -year flood. By weighting predictions based on model performance, BMA offers a structured way to combine multiple estimates, helping to improve overall predictive robustness. However, inter-

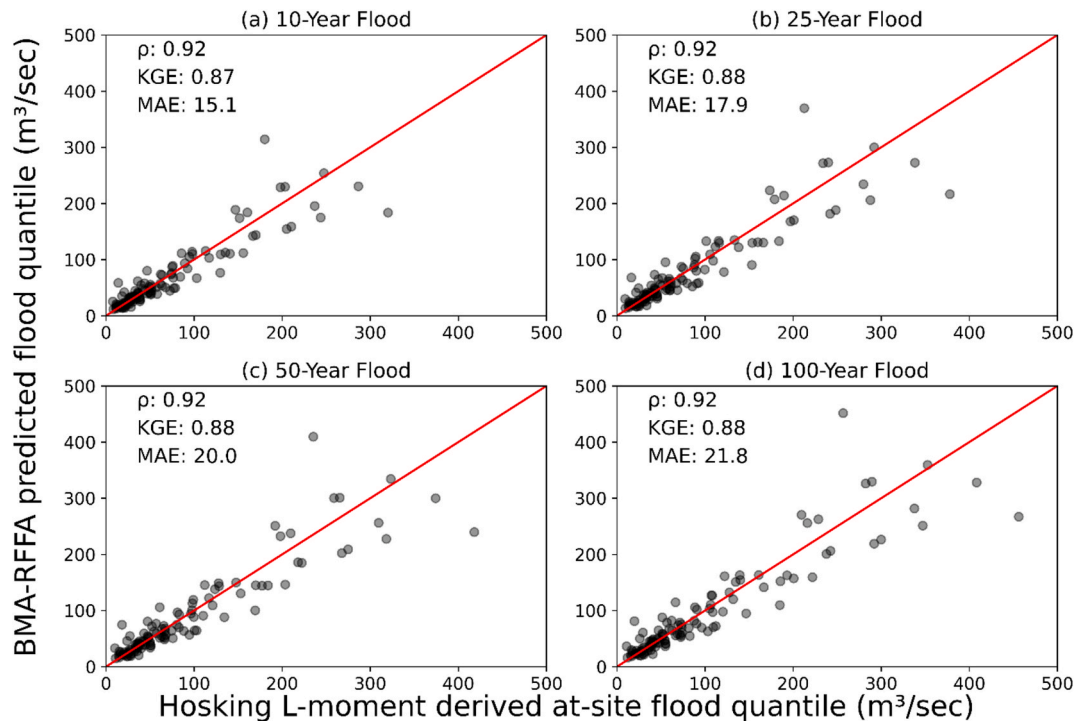


Fig. 6. Comparison between Hosking L-moment derived at-site flood quantile (used as response variable in regression model development) and BMA-RFFA model predicted flood quantiles for (a) 10-year, (b) 25-year, (c) 50-year, and (d) 100-year Floods using LOOCV evaluation approach. Climate predictors extracted from ERA5 data are used for this model development.

model predictive uncertainty among regression models remains, particularly for higher return periods (shown in Fig. S7), which should be considered when interpreting the results.

4.4. Predictor importance and model sensitivity evaluation

To visualize the influence of input variables on flood quantile prediction, the SHAP method is applied to all five regression models across multiple return periods (10-year, 25-year, 50-year and 100-year). Among the models, the RF model at the 100-year return period is selected for detailed discussion due to the prominence of the 100-year flood in infrastructure design and the model's inherent interpretability in feature attribution. The variation in SHAP values across return periods was found to be less than the variation across different models; hence, only model-wise comparisons are included in the [Supplementary Material \(Fig. S9\)](#) to maintain focus. While all five individual models demonstrate satisfactory performance in predicting flood quantiles, the RF model is selected to examine variable importance due to its built-in capability to assess variable importance, quantifying the contribution of predictors to the response variable. Results show that catchment area is the most important factor in predicting flood quantile, followed by seasonal R10, seasonal water storage, slope and 100-year rainfall (Fig. 7). A closely similar ranking is found using the RF's inherent variable importance metric, with catchment area, seasonal R10, seasonal water storage, 100-year rainfall, and slope identified as the top five contributors (Fig. S10). While the contribution of drainage area is substantially higher than that of any other predictor, the inclusion of time-varying climatic variables, ranked highly, is also critical, particularly for predicting flood quantiles under different warming scenarios. A key advantage of using SHAP method over RF variable importance method, it illustrates the positive and negative contributions of each variable, along with their impact on individual predictions. For instance, catchment area, seasonal water storage predominantly contributes positively, while main channel slope and lake wetland (%) exhibit negative associations. These findings accurately reflect the functioning of the hydrologic system. Positive associations indicate that larger values of these variables generally result in higher flood quantile values, while negative associations suggest a decrease in flood quantiles with increasing variable values. The mixed contribution of seasonal R10 suggest a context-dependent influence on flood response. In some catchments, higher seasonal R10 values, indicating more frequent but typically lower-intensity rainfall, may reduce flood peaks by distributing precipitation

more evenly and limiting short-duration extremes. This effect is especially evident when such rainfall does not align with high-intensity storm events or snowmelt periods. Conversely, in other catchments, frequent moderate rainfall may precondition soils and elevate antecedent moisture levels, thereby amplifying runoff generation during subsequent intense storms. This dual role is consistent with established hydrologic understanding (Merz and Blöschl, 2009; Singh N. K. et al., 2021), particularly in transitional or snow-influenced catchments where the timing, intensity, and sequence of rainfall events interact to shape flood responses.

Fig. 8 shows the relationship between catchment area and model performance by plotting the bias range (10th–90th percentile) against various catchment area quantiles. The results reveal that the BMA-RFFA model exhibits increased variability in prediction bias for catchments in the higher quantiles of drainage area. While no formal statistical correlation was tested, this pattern may be partly explained by the fact that catchment area is the primary predictor of flood quantiles and often maintains a positive relationship with flood magnitude in humid and temperate regions, larger catchment areas are generally associated with higher flood quantiles in the study area, which in turn can lead to increased prediction bias. However, this elevated bias diminishes when

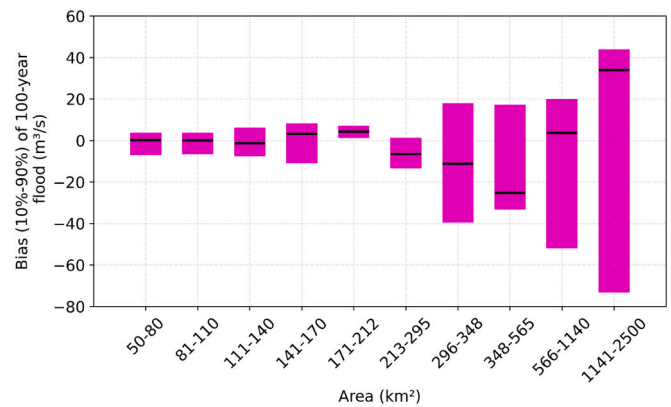


Fig. 8. Bias range (10th–90th percentile) of predicted 100-year flood quantiles from the BMA-RFFA model across catchment area percentiles (10th, 20th, 30th, 40th, 50th, 60th, 70th, 80th, 90th, and 99th). The black line represents the median of the bias range.

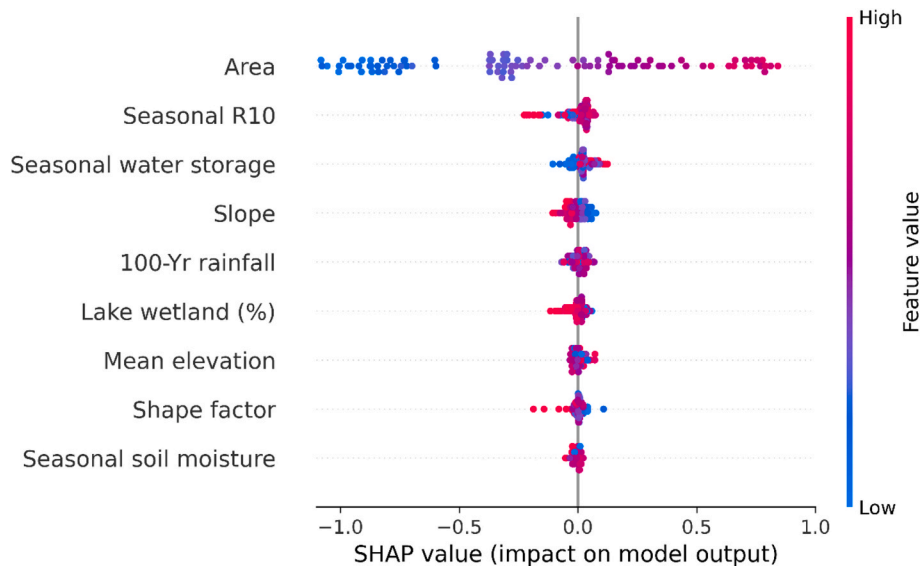


Fig. 7. SHAP Summary Plot: The y-axis displays feature ranking based on the importance within the prediction model. Each dot represents a sample, with its x-position indicating the impact of the corresponding predictor variable on the predicted 100-year flood quantile using the RF regression model.

expressed as relative bias (%) (Fig. S11). Additionally, the balanced distribution of bias in both positive and negative directions supports key regression assumptions, such as homoscedasticity.

4.5. Projected changes in design flood under different warming levels

Changes in flood quantiles for studied gauge stations in southern Ontario are assessed using the BMA-RFFA model, with time-varying predictors derived from the CanRCM4-LE climate ensemble. The projected changes in flood quantiles are attributed to changes in climatic variables including seasonal water storage, seasonal R10, rainfall quantiles, and seasonal soil moisture. Future flood quantiles are estimated by applying the calibrated BMA-RFFA model using climatic predictors extracted from ensemble members under different warming levels, as described in the methodology section.

Fig. 9 illustrates the projected changes in the 100-year flood quantile under three different global warming scenarios: +1.5 °C (2024–2054), +2 °C (2034–2064), and +3 °C (2053–2083), along with the associated ensemble range (5 %, median, and 95 %) conditioned on a single model structure and emission scenario. Similar spatial patterns are observed for other flood quantiles, including the 10-year, 25-year, and 50-year events, as shown in Fig. S12–S14. The results indicate that the projected changes in the 100-year flood quantiles range from −19 % to +43 % for different catchments across ensemble members (5 %–95 %). The range highlights the inherent ensemble spread in flood projections, driven by varying initial conditions and internal variability within the regional climate model.

The projected median 100-year flood changes across the stations range from −13 % to +20 % under different warming scenarios. The results reveal a near-even distribution of change directions, with 58, 60, and 59 out of 114 stations showing positive changes under the 1.5 °C, 2.0 °C, and 3.0 °C warming levels respectively (Fig. S15). While spatial clusters of positive change are observed in the southern and western portions of the study area, some catchments, particularly those located

in the inland area between Lake Huron and Lake Erie, exhibit negative trends. These patterns may be influenced by region-specific hydrometeorological processes, including lake-induced microclimates, altered snowfall and snowmelt regimes, and shifts in antecedent soil moisture conditions under warming scenarios. In transitional basins, reduced snow accumulation and earlier melt timing can decouple peak runoff from heavy precipitation events, potentially dampening spring flood magnitudes despite overall climatic warming (Poitras et al., 2011; Vincent et al., 2015). This spatial variability underscores the complex response of flood behavior to future climate change across diverse catchment characteristics in the region.

Fig. 10 illustrates the projected ensemble median changes in climatic predictors extracted from the CanRCM4-LE model for 114 catchments. The figure presents anticipated changes in seasonal water storage, R10 (the number of days with rainfall exceeding 10 mm), rainfall intensity (100-year rainfall), and seasonal soil moisture across three different warming levels: +1.5 °C (2024–2054), +2 °C (2034–2064), and +3 °C (2053–2083), compared to a baseline period from 1986 to 2016.

The results indicate that seasonal soil moisture, rainfall intensity, and R10 are projected to increase across all three warming levels, with the intensity of these increases becoming more pronounced as the warming level rises from +1.5 °C to +3.0 °C. In contrast, seasonal water storage is expected to decrease at all warming levels, with a more significant decline observed at higher temperatures. This reduction in seasonal water storage is attributed to the relationship defined in Equation (11), which incorporates seasonal total rainfall, PET, and SWE. Although seasonal rainfall is anticipated to increase, the decline in seasonal SWE and the rise in PET are expected to occur at a higher rate under elevated warming scenarios. As a result, the total water storage is expected to decrease. Seasonal water storage is projected to decline significantly in catchments along Lake Ontario and in the central region, where decreases are expected to be more pronounced than in other areas. Seasonal R10 and seasonal soil moisture exhibit a similar spatial pattern, with lower increases observed in catchments near Lake Huron,

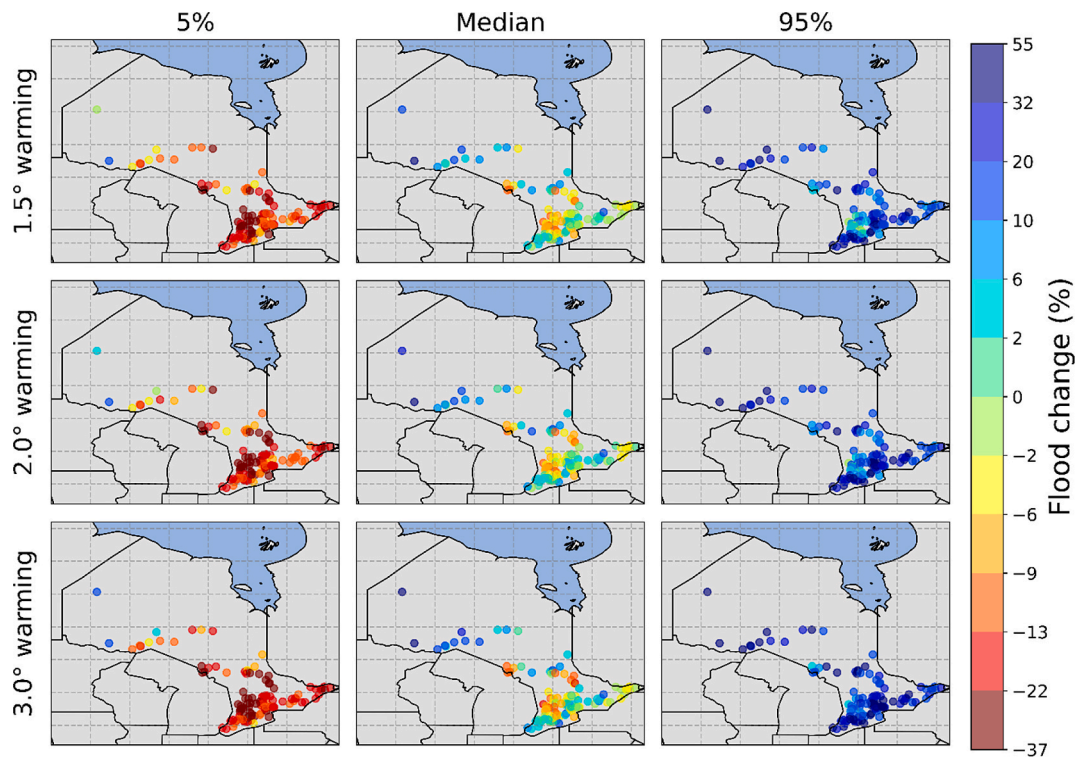


Fig. 9. Projected ensemble range (5th percentile, median, and 95th percentile) for changes in 100-year flood quantiles under three different warming scenarios: +1.5 °C (2024–2054), +2 °C (2034–2064), and +3 °C (2053–2083) relative to the baseline period (1986–2016) for 114 catchments across the Great Lakes region in Ontario. Flood quantiles are estimated using the BMA-RFFA model with climatic predictors derived from CanRCM4-LE ensemble members.

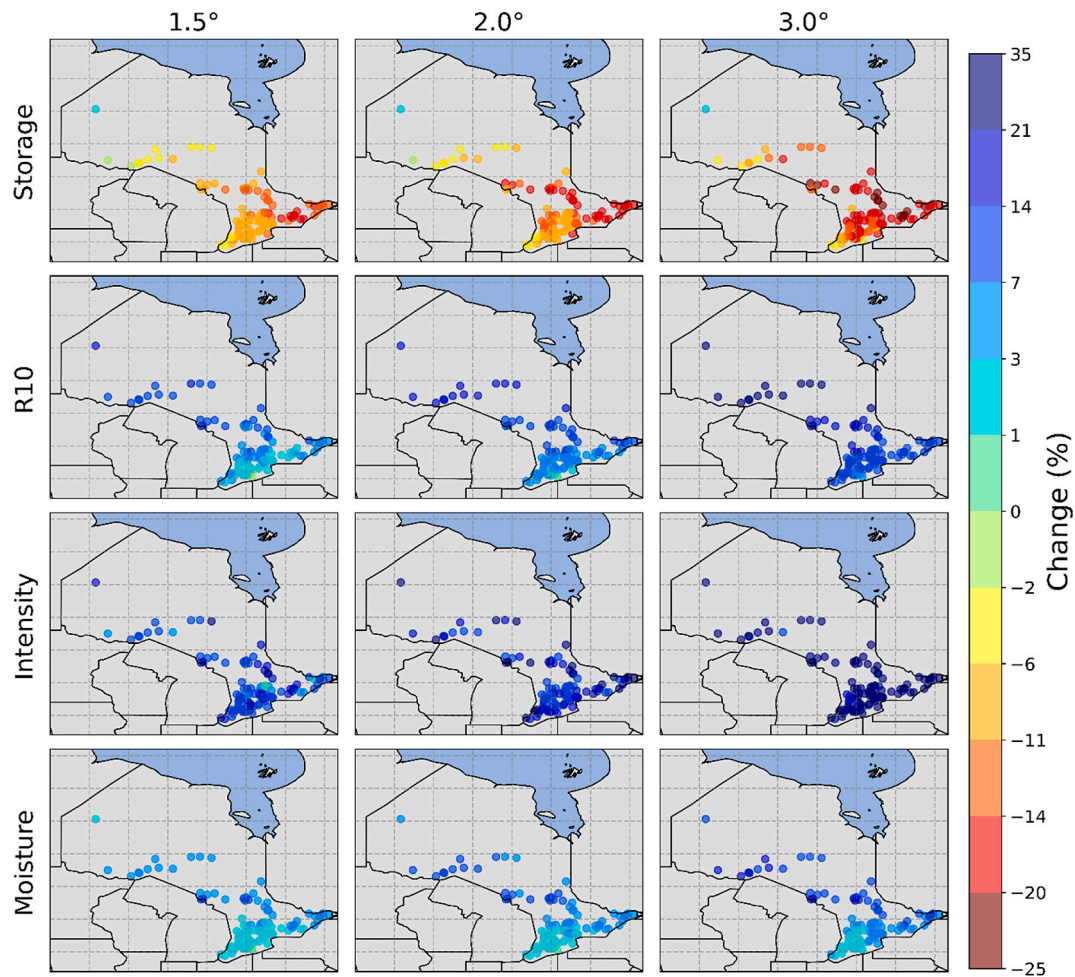


Fig. 10. Projected ensemble median changes in climatic predictors extracted from CanRCM4-LE: seasonal water storage, seasonal R10, rainfall intensity, and seasonal soil moisture under three different warming levels (+1.5 °C [2024–2054], +2°C [2034–2064], and +3°C [2053–2083]) compared to the base period (1986–2016) for 114 catchments.

Lake Erie, and the southern parts of Lake Ontario. In contrast, catchments situated further away from these lakes, especially in the upper latitudes, display a more pronounced increasing trend. However, rainfall intensity does not exhibit a distinct spatial pattern. Additionally, variations in wind patterns and atmospheric circulation may influence precipitation distribution, snowmelt, and temperature, further contributing to the observed differences in flood quantile changes. This complex interplay among climatic variables is critical in shaping the spring flood generation process, as defined in this study. These spatial patterns may be influenced by lake effects and broader atmospheric circulation changes that affect precipitation, snowmelt, and temperature. While CanRCM4 incorporates lake–atmosphere interactions through a coupled lake model, localized lake-effect processes may not be fully resolved at the model's spatial resolution. Our interpretation is guided by model outputs and existing literature; however, a formal process-based evaluation of these mechanisms is beyond the scope of this study. Future work involving physically based modeling and deductive analysis would be valuable to validate these interpretations.

The projected median changes in flood quantiles identified in this study partly differ from the findings of [Gizaw and Gan \(2016\)](#). Their research focused on 23 catchments in Southern Ontario and utilized a limited set of climatic predictors, primarily analyzing annual floods rather than specifically examining spring floods. Their exclusive consideration of rainfall as a predictor led to higher projected increases in flood magnitudes. In contrast, our analysis incorporates additional hydroclimatic drivers relevant to spring floods, including SWE and PET.

A significant decline in SWE and a projected rise in PET under future warming are expected to reduce antecedent snow accumulation and increase seasonal moisture losses. These processes may offset the effects of increased seasonal rainfall frequency and intensity, thereby moderating the resulting flood quantile.

It is essential to acknowledge the inherent uncertainties and limitations associated with climate projections. Further research is needed to gain deeper insights into the underlying drivers of the projected changes in flood quantiles across the studied region. However, this study contributes to the existing body of knowledge by presenting the range of flood quantile changes under different warming levels projected by multiple ensemble members, an aspect that has received limited attention in previous regional studies. It is also important to note that this analysis operates under the assumption that only climatic variables will change over time, while physiographic characteristics such as basin area, mean elevation, channel shape, and slope are considered constant. These characteristics are generally stable over decadal time periods particularly in natural basins; however, we acknowledge that changes due to land use or river engineering may occur in some catchments, which are not accounted for in this assessment. These aspects represent important avenues for future research.

4.6. Temporal evolution of flood quantiles

The temporal evolution of flood quantiles is illustrated using ensemble pooling approach for each 10-year block over the period from

1950 to 2100. Fig. 11 displays the continuous temporal evolution of median flood quantiles over 50 ensemble members across all 114 stations in the study area. The observed trends are consistent across different return periods (e.g., 10, 25, 50, and 100 years). Changes are presented relative to the base period (1986–2016). Most stations demonstrate a gradual increase in flood quantiles up until 2005, with relatively narrow variability. The observed dispersion patterns in the later decades reflect increased variability in climatic drivers, contributing to temporal evolution of flood responses among the stations. These findings highlight how flood risk is expected to evolve over time and provide insights into potential future hydrological conditions.

Fig. 12 illustrates the spatial variability in projected changes in median flood quantiles using ensemble pooling approach for the 10-year, 25-year, 50-year, and 100-year floods across the study area for the last 10-year block (2091–2100) relative to the base period (1986–2016). This figure essentially provides a spatial representation of the patterns shown in Fig. 11 but focuses exclusively on the final decade. The projected changing patterns align with the median flood changes shown in Fig. 9; however, the magnitude of change is more pronounced by the end of the century compared to different warming levels shown in Fig. 9. Results indicate that stations on the western side, along with those near Lake Erie and Lake Ontario, are expected to experience predominantly positive changes in flood magnitudes. In contrast, stations in the far eastern region and many located in the central areas, particularly around Lake Huron, Lake Erie, and Lake Ontario, show relatively minor changes, with most showing of negative trends. This reflects localized variability in flood responses. Notably, changes are relatively pronounced for higher flood quantiles, especially the 100-year and 50-year floods, highlighting their greater sensitivity to projected climatic shifts. This observed spatial heterogeneity underscores the varying impacts of climate change on flood risks across different catchments, likely influenced by local climatic shifts, catchment

physiography, and lake effects that may moderate or amplify flood responses in certain regions.

Even though climate model projections introduce significant uncertainty, the findings of this study advocate for the utilization of single model initial-condition large ensembles based on pooling approach. This method allows for statistically consistent modeling over time by augmenting sample size and enables the exploration of flood temporal evolution by examining the range of projected flood quantiles across ensemble members over time.

4.7. Relative contribution of ACC and ICV on future flood projections

Fig. 13 presents the spatial distribution of anthropogenic climate change (ACC) impacts on 100-year flood magnitudes across 114 catchments in southern Ontario under three different warming levels relative to the baseline period (1986–2016). These projections, based on the ensemble mean of CanRCM4-LE, capture the collective response of the model ensemble to external forcings. Positive values represent an increase in flood magnitudes due to ACC, while negative values indicate a decrease. The magnitude of these changes, whether positive or negative, reflects the relative influence of ACC on future design flood estimates.

The spatial pattern of ACC impacts shows general consistency across warming levels, with slight amplification at +3°C. Stations near the Great Lakes predominantly exhibit a positive response to ACC, while catchments in the interior regions, particularly those in the central parts of Lake Huron and Lake Erie, show a negative response. This spatial distribution aligns closely with the median flood change pattern shown in Fig. 9, suggesting spatial coherence and internal consistency in the projected ACC effects across different warming levels and return periods. The relative ACC impacts range from –22 % to +25 % across different warming levels, with most stations experiencing changes within –10 % to +12 %.

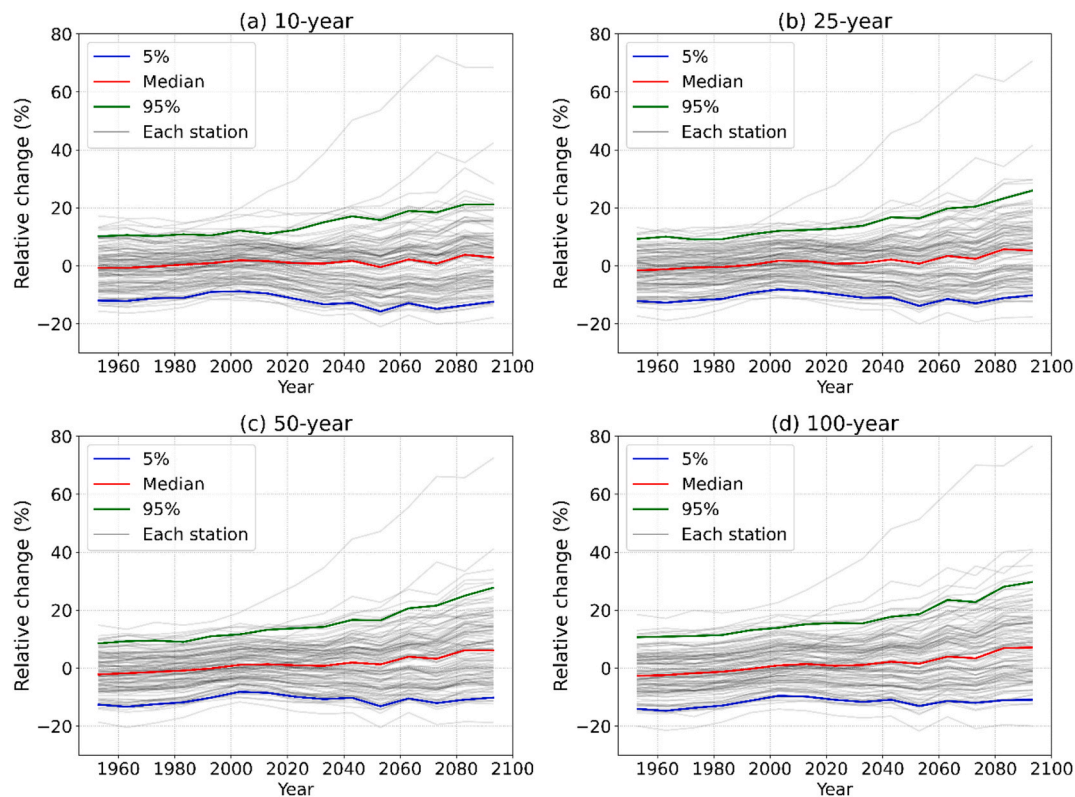


Fig. 11. Continuous relative change trends of various flood quantiles: (a) 10-year, (b) 25-year, (c) 50-year, and (d) 100-year flood, analyzed at 10-year intervals from 1951 to 2100 using the ensemble pooling approach, relative to the base period (1986–2016). The black lines indicate the median change trends over 50 ensembles for each of the 114 stations studied, while the blue, red, and green lines represent the 5th percentile, median, and 95th percentile change scenarios derived from the aggregated trends across all stations. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

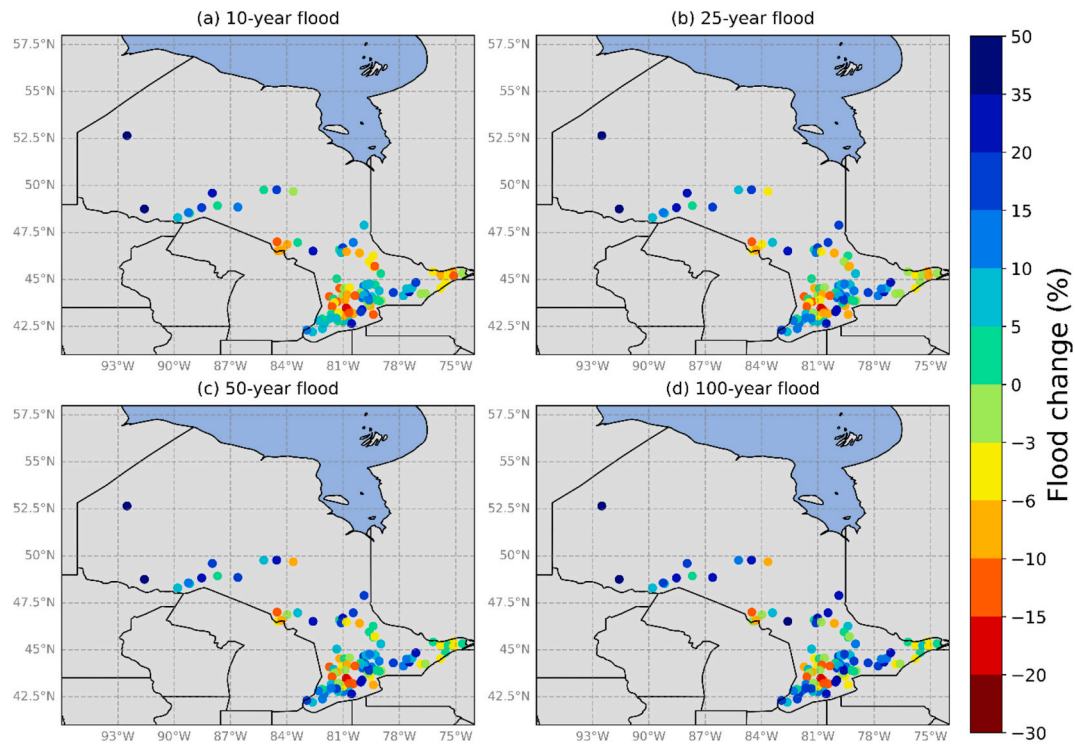


Fig. 12. Spatial distribution of projected median changes in various flood quantiles: (a) 10-year, (b) 25-year, (c) 50-year, and (d) 100-year flood, derived using the ensemble pooling approach for the last decade of the century (2091–2100) relative to the base period (1986–2016).

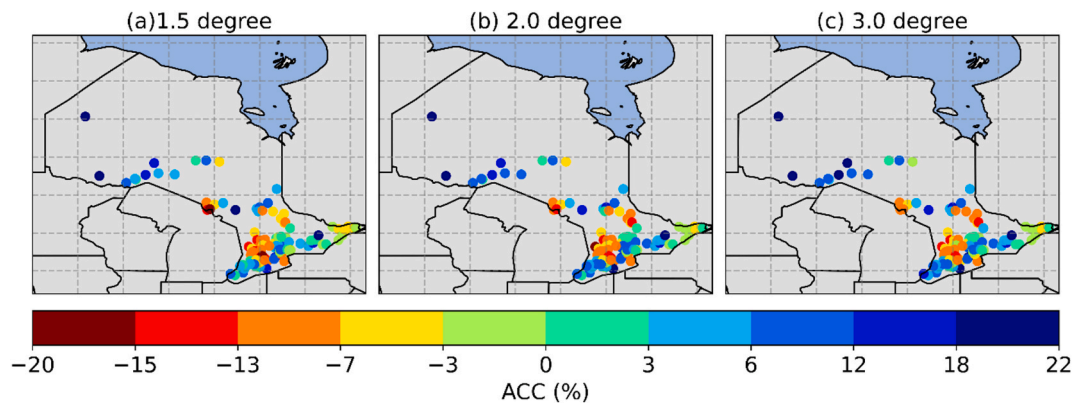


Fig. 13. Projected impact of anthropogenic climate change (ACC) on the 100-year flood quantile. The ACC impact is represented as the relative change (%) and is estimated using the BMA-RFFA model for three distinct warming levels (a) +1.5 °C (2024–2054), (b) +2 °C (2034–2064), and (c) +3 °C (2053–2083) compared to the base period (1986–2016) for 114 catchments. Climatic predictors are derived from the CanRCM4-LE ensemble members, and ACC is calculated as the mean percentage change across all ensemble members.

Fig. 14 illustrates the influence of internal climate variability (ICV) on projected changes in 100-year flood magnitudes for different warming levels, reflecting the spread among the CanRCM4-LE ensemble members. The spatial patterns are consistent across the warming scenarios, but the ICV response becomes more pronounced at the highest warming level (+3°C). A higher percentage of stations located closer to the Great Lakes exhibit comparatively greater ICV response, suggesting that local climatic conditions influenced by the lake effect significantly contribute to the variability in flood projections. The higher magnitude of ICV than ACC for any station indicates the role of anthropogenic influence may be offset by the increased noise resulting from internal climate variability. These results emphasize the complex interplay between ACC and ICV in shaping future flood risk. This aligns with the findings of [Faghih and Brissette \(2023\)](#), which were conducted in the northern part of the study area, and highlights the importance of

considering both factors when evaluating long-term flood risk and developing effective adaptation strategies.

The contrasting patterns of ACC and ICV influence underscore the importance of tailoring adaptation strategies: regions with a stronger ACC signal, such as those in the western part, are likely to face flood risks primarily driven by human-induced changes, necessitating long-term infrastructure adaptation strategies. In contrast, the majority of stations dominated by ICV, including catchments in the central part, will likely need to prioritize flexible management approaches that account for short-term fluctuations in flood risks. Stations closer to the Great Lakes exhibit higher ICV responses, with variability magnitudes surpassing those of ACC, suggesting that natural variability plays a more significant role in shaping flood risk projections in these areas. This dual dominance of ACC and ICV highlights the need for region-specific flood risk management strategies, balancing infrastructure adaptation with

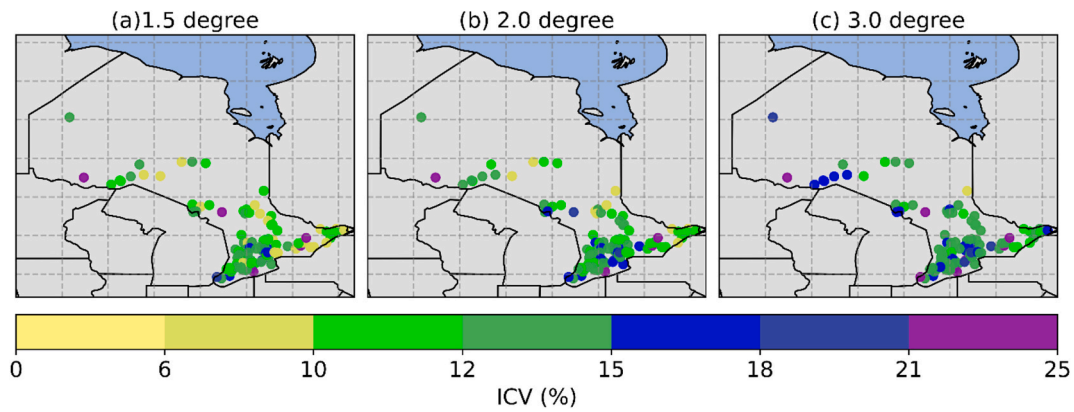


Fig. 14. Projected internal climate variability (ICV) response for the 100-year flood across 114 catchments for three different warming levels (a) +1.5 °C (2024–2054), (b) +2 °C (2034–2064), and (c) +3 °C (2053–2083) relative to the base period (1986–2016). The relative ICV response (%) is derived using the BMA-RFFA model based on climatic predictors from the CanRCM4-LE ensemble members. ICV is estimated as the standard deviation from the mean of the ensemble's percentile changes.

flexible, scenario-based approaches.

Fig. 15 presents a detailed analysis of Signal-to-Noise Ratio (SNR) values for 100-year floods across different warming levels, illustrating the relative contributions of ACC and ICV to future flood changes. SNR values are crucial metrics: values above 1.0 or below -1.0 indicate that ICV plays a diminished role compared to ACC, while values between -1.0 and $+1.0$ suggest that ICV has a more significant impact. The sign of the SNR further indicates the projected direction of flood changes.

The dominance of ICV is notably pronounced across most catchments under various warming levels, except in the western regions where a positive signal predominates over the noise introduced by ICV. The spatial patterns of SNR remain consistent across different warming levels, with ACC's influence becoming more evident in few additional stations under the +3 °C scenario. However, the underlying causes of these patterns warrant further investigation, presenting promising opportunities for future research.

Additionally, the spatial distribution of basins in Fig. 15 highlights that as warming levels increase, the contributions of ACC and ICV become more pronounced shown in Fig. 13 and Fig. 14 respectively, reflecting a consistent trend in SNR values across different warming scenarios. Overall, these results provide preliminary insights into the relative roles of ACC and ICV in projected flood quantiles under varying warming levels, though the substantial uncertainties inherent in climate projections necessitate cautious interpretation. Further research is needed to refine these estimates and explore the mechanisms driving the observed spatial patterns.

5. Conclusions

While traditional flood estimation methods relying on historical climatic data and stationarity assumptions remain foundational, their application under changing conditions requires careful consideration of long-term climate variability, which can confound trend attribution. New strategies are required to develop methodological framework that explicitly address dynamic hydrological processes and incorporate changing flood behavior under projected climate variability to better represent evolving flood risks under climate change. To address this challenge, this study develops a multi-model RFFA framework using BMA to evaluate the projected changes in spring flood quantiles under different warming scenarios (+1.5 °C, +2 °C, and +3 °C) for 114 hydrologically homogenous catchments in Southern Ontario, along the Great Lakes in Canada. Approximately 80 % of the floods in this region occur during the spring, making it a critical season for flood risk assessment. The analysis focused on spring floods due to their complex flood generation mechanisms, which are governed by the interplay of SWE, rainfall, and PET. The BMA-RFFA model demonstrates consistent predictive skill in both spatial and temporal validation approaches, outperforming individual regression models across most evaluation metrics. For instance, KGE and Spearman rank correlation coefficient (ρ) values exceed 0.83 and 0.90, respectively, across various flood quantiles, in predicting the response variable either derived from the regionalized frequency analysis or empirical estimates. Time-varying climatic predictors extracted from the CanRCM4-LE ensemble dataset are used to predict future flood quantiles, assess temporal evolution of

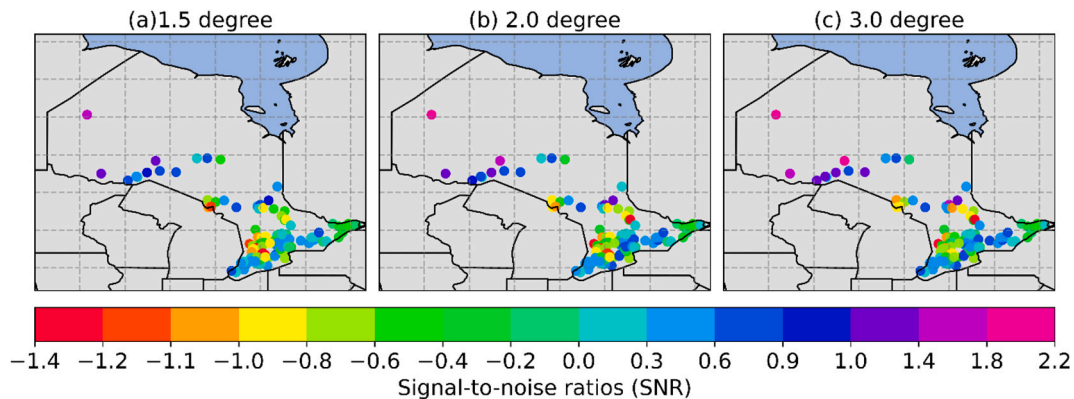


Fig. 15. Signal-to-Noise Ratios (SNR) for projected 100-year design floods across 114 catchments at three warming levels: (a) +1.5 °C (2024–2054), (b) +2 °C (2034–2064), and (c) +3 °C (2053–2083) relative to the base period (1986–2016). SNR is calculated as the ratio of ACC to ICV.

flood quantiles driven by climate ensemble variability, and evaluate the relative contributions of ACC and ICV on projected flood magnitudes. The key findings of this study are given below:

1. Projections indicate a general increase in spring flood quantiles across most of higher and lower latitude catchments, with median flood quantiles increasing by up to 20 % under the +3°C scenario. However, catchments in the central region generally exhibit a decrease in flood quantiles, highlighting spatial variability in flood responses. Although the overall median change remains modest, the magnitude of change becomes more pronounced as warming levels intensify from +1.5 °C to +3°C.
2. The study incorporates a 50-member ensemble dataset to capture the range of projected flood responses, revealing a substantial spread, especially for high-return period floods (e.g., 100-year floods), with notable spatial variation across the study area. This wide ensemble range underscores the need to consider internal variability when assessing future flood risks, as neglecting this spread could lead to underestimation or overestimation of potential flood magnitudes.
3. The inclusion of a broader set of climatic variables such as seasonal water storage (combining seasonal rainfall, PET and SWE), seasonal R10 (number of days with rainfall >10 mm), and seasonal soil moisture proved crucial in explaining projected future flood dynamics. To the authors' knowledge, these variables have not been jointly considered in previous studies for this region, and their inclusion offers new insights into the climate-driven mechanisms influencing future flood behavior. The results show that decreases in seasonal water storage combined with increases in rainfall intensity and soil moisture are primary drivers of projected flood changes under higher warming levels.
4. The temporal evolution of projected flood quantiles is examined using ensemble pooling approach. Results show a relatively consistent pattern of change across stations until 2005, after which variability across catchments increases, with a more distinct spread by the end of the century. This growing temporal divergence suggests a shift in flood behavior over time, necessitating dynamic approaches for future flood estimation.
5. The influence of ACC and ICV varies spatially across the region. The SNR analysis shows that ICV is dominant in most catchments prioritizing flexible management approaches under various warming levels, except in the western regions where ACC shows a stronger influence necessitating long-term infrastructure adaptation strategy. The responses of ACC and ICV become more pronounced under higher warming levels; however, the intensification is more prominent for ICV across most stations, except for those in higher-latitude regions of the western part, where ACC exhibits a stronger influence.

While this study incorporates a broad set of hydroclimatically relevant predictors, future research could further refine regional models by exploring additional physically based or dynamically evolving variables to enhance explanatory power without increasing model complexity. Additionally, investigating the influence of broader land use changes, including vegetation dynamics and urbanization, may offer further insights into long-term flood behavior. Expanding the analysis to other seasons and extreme weather events would also provide a more holistic view of future flood risks. Furthermore, employing advanced convective-permitted fine resolution ensemble modeling techniques to quantify nonstationary flood risks and explore mitigation strategies will be essential for designing resilient flood management plans in the context of evolving climate patterns. Such efforts are critical for improving long-term flood risk assessments and ensuring infrastructure resilience in regions with complex hydrological systems, such as those around the Great Lakes.

CRedit authorship contribution statement

Md. Robiul Islam: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mohammad Reza Najafi:** Writing – review & editing, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhydrol.2025.133703>.

Data availability

Data will be made available on request.

References

- AghaKouchak, A., Easterling, D., Hsu, K., Schubert, S., Sorooshian, S., 2012. *Extremes in a changing climate: detection, analysis and uncertainty*. Springer Science & Business Media.
- Ahn, K.-H., Palmer, R., 2016. Regional flood frequency analysis using spatial proximity and basin characteristics: quantile regression vs. parameter regression technique. *J. Hydrol.* 540, 515–526. <https://doi.org/10.1016/j.jhydrol.2016.06.047>.
- Allahbakhshian-Farsani, P., Vafakhah, M., Khosravi-Farsani, H., Hertig, E., 2020. Regional flood frequency analysis through some machine learning models in semi-arid regions. *Water Resour. Manag.* 34, 2887–2909. <https://doi.org/10.1007/s11269-020-02589-2>.
- de Almeida, I.K., Almeida, A.K., Anache, J.A.A., Steffen, J.L., Sobrinho, T.A., 2014. Estimation on time of concentration of overland flow in watersheds: a review. *Geosci. Geociências* 33, 661–671.
- Awasthi, C., Archfield, S.A., Ryberg, K.R., Kiang, J.E., Sankarasubramanian, A., 2022. Projecting flood frequency curves under near-term climate change. *Water Resour. Res.* 58, e2021WR031246. <https://doi.org/10.1029/2021WR031246>.
- Aziz, K., Rahman, A., Fang, G., Shrestha, S., 2014. Application of artificial neural networks in regional flood frequency analysis: a case study for Australia. *Stoch. Environ. Res. Risk Assess.* 28, 541–554. <https://doi.org/10.1007/s00477-013-0771-5>.
- Bardsley, W.E., Vetrova, V., Liu, S., 2015. Toward creating simpler hydrological models: a LASSO subset selection approach. *Environ. Model. Softw.* 72, 33–43.
- Ben Alaya, M.A., Zwiers, F., Zhang, X., 2020. An evaluation of block-maximum-based estimation of very long return period precipitation extremes with a large ensemble climate simulation. *J. Clim.* 33, 6957–6970.
- Berghuijs, W.R., Harrigan, S., Molnar, P., Slater, L.J., Kirchner, J.W., 2019. The relative importance of different flood-generating mechanisms across Europe. *Water Resources Research* 55 (6), 4582–4593. <https://doi.org/10.1029/2019WR024841>.
- Boni, G., Ferraris, L., Giannoni, F., Roth, G., Rudari, R., 2007. Flood probability analysis for un-gauged watersheds by means of a simple distributed hydrologic model. *Adv. Water Resour. Recent Dev. Hydrol. Anal.* 30, 2135–2144. <https://doi.org/10.1016/j.advwatres.2006.08.009>.

- Bracken, C., Holman, K.D., Rajagopalan, B., Moradkhani, H., 2018. A Bayesian hierarchical approach to multivariate nonstationary hydrologic frequency analysis. *Water Resour. Res.* 54, 243–255. <https://doi.org/10.1002/2017WR020403>.
- Buishand, T.A., 1982. Some methods for testing the homogeneity of rainfall records. *J. Hydrol.* 58, 11–27.
- Burn, D.H., 1990. Evaluation of regional flood frequency analysis with a region of influence approach. *Water Resour. Res.* 26, 2257–2265.
- Buttle, J.M., Allen, D.M., Caissie, D., Davison, B., Hayashi, M., Peters, D.L., Pomeroy, J. W., Simonovic, S., St-Hilaire, A., Whitfield, P.H., 2016. Flood processes in Canada: regional and special aspects. *Can. Water Resour. Journal/revue Can. Ressour. Hydr.* 41, 7–30.
- Byun, K., Hamlet, A.F., 2020. A risk-based analytical framework for quantifying non-stationary flood risks and establishing infrastructure design standards in a changing environment. *J. Hydrol.* 584, 124575.
- Cannon, A.J., Jeong, D.I., Zhang, X., Zwiers, F.W., 2020. Climate-resilient buildings and core public infrastructure. *Assess. Impact Clim. Change Clim. Des. Data Can. Environ. Clim. Change Can.*, 106.
- CDA, 2013. Dam Safety Guidelines.
- Cheng, L., AghaKouchak, A., 2014. Nonstationary precipitation intensity-duration-frequency curves for infrastructure design in a changing climate. *Sci. Rep.* 4, 1–6.
- Chow, V.T., Maidment, D.K., Mays, L.W., 1988. *Applied Hydrology*. McGraw-Hill Book Company, N. Y.
- Dai, A., Bloecker, C.E., 2019. Impacts of internal variability on temperature and precipitation trends in large ensemble simulations by two climate models. *Clim. Dyn.* 52, 289–306. <https://doi.org/10.1007/s00382-018-4132-4>.
- Dalrymple, T., 1960. Flood-frequency analysis: US Geological Survey Water-Supply Paper 1543-A. *Man. Hydrol.* Pt 3.
- Delworth, T., Manabe, S., Stouffer, R.J., 1993. Interdecadal variations of the thermohaline circulation in a coupled ocean-atmosphere model. *J. Clim.* 6, 1993–2011. [https://doi.org/10.1175/1520-0442\(1993\)006<1993:IVOTTC>2.0.CO;2](https://doi.org/10.1175/1520-0442(1993)006<1993:IVOTTC>2.0.CO;2).
- Desai, S., Ouara, T.B.M.J., 2021. Regional hydrological frequency analysis at ungauged sites with random forest regression. *J. Hydrol.* 594, 125861. <https://doi.org/10.1016/j.jhydrol.2020.125861>.
- Deser, C., 2020. Certain uncertainty: the role of internal climate variability in projections of regional climate change and risk management. *Earths Future* 8, e2020EF001854. <https://doi.org/10.1029/2020EF001854>.
- Deser, C., Knutti, R., Solomon, S., Phillips, A.S., 2012. Communication of the role of natural variability in future North American climate. *Nat. Clim. Change* 2, 775–779.
- Deser, C., Terray, L., Phillips, A.S., 2016. Forced and internal components of winter air temperature trends over North America during the past 50 years: mechanisms and implications. *J. Clim.* 29, 2237–2258. <https://doi.org/10.1175/JCLI-D-15-0304.1>.
- Dikshit, A., Pradhan, B., 2021. Interpretable and explainable AI (XAI) model for spatial drought prediction. *Sci. Total Environ.* 801, 149797.
- Duan, Q., Ajami, N.K., Gao, X., Sorooshian, S., 2007. Multi-model ensemble hydrologic prediction using Bayesian model averaging. *Adv. Water Resour.* 30, 1371–1386.
- Dunne, T., Leopold, L.B., 1978. *Water in environmental planning*. Macmillan.
- Durocher, M., Chebana, F., Ouara, T.B., 2015. A nonlinear approach to regional flood frequency analysis using projection pursuit regression. *J. Hydrometeorol.* 16, 1561–1574.
- Durocher, M., Chebana, F., Ouara, T.B.M.J., 2016. Delineation of homogenous regions using hydrological variables predicted by projection pursuit regression. *Hydrol. Earth Syst. Sci.* 20, 4717–4729. <https://doi.org/10.5194/hess-20-4717-2016>.
- El Adlouni, S., Ouara, T.B.M.J., Zhang, X., Roy, R., Bobée, B., 2007. Generalized maximum likelihood estimators for the nonstationary generalized extreme value model. *Water Resour. Res.* 43. <https://doi.org/10.1029/2005WR004545>.
- England, J.F., Cohn, T.A., Faber, B.A., Stedinger, J.R., Jr, W.O.T., Veilleux, A.G., Kiang, J. E., Robert R. Mason, J., 2019. Guidelines for determining flood flow frequency — Bulletin 17C (No. 4-B5), Techniques and Methods. U.S. Geological Survey. <https://doi.org/10.3133/tm4B5>.
- Faghih, M., Brissette, F., 2023. The role of internal climate variability on future streamflow projections. *J. Hydrol.*, 130101.
- Feldstein, S.B., 2000. The timescale, power spectra, and climate noise properties of teleconnection patterns. *J. Clim.* 13, 4430–4440. [https://doi.org/10.1175/1520-0442\(2000\)013<4430:TTPSAC>2.0.CO;2](https://doi.org/10.1175/1520-0442(2000)013<4430:TTPSAC>2.0.CO;2).
- Fereshtehpour, M., Najafi, M.R., 2025. Urban stormwater resilience: Global insights and strategies for climate adaptation. *Urban Clim.* 59, 102290.
- Fereshtehpour, M., Najafi, M.R., Leach, J.A., Wang, Y., 2025. Quantifying the individual and combined influence of climate change, land cover transition, and internal climate variability on the hydrology of a snow-dominated forested watershed. *Climatic Change* 178 (2), 1–32. <https://doi.org/10.1007/s10584-024-03848-6>.
- Freijeiro-González, L., Febrero-Bande, M., González-Manteiga, W., 2022. A critical review of LASSO and its derivatives for variable selection under dependence among covariates. *Int. Stat. Rev.* 90, 118–145. <https://doi.org/10.1111/insr.12469>.
- Gaetani, M., Janicot, S., Vrac, M., Famién, A.M., Sultan, B., 2020. Robust assessment of the time of emergence of precipitation change in West Africa. *Sci. Rep.* 10, 7670. <https://doi.org/10.1038/s41598-020-63782-2>.
- Gao, Z., Long, D., Tang, G., Zeng, C., Huang, J., Hong, Y., 2017. Assessing the potential of satellite-based precipitation estimates for flood frequency analysis in ungauged or poorly gauged tributaries of China's Yangtze River basin. *J. Hydrol.* 550, 478–496. <https://doi.org/10.1016/j.jhydrol.2017.05.025>.
- Gillett, N.P., Cannon, A.J., Malinina, E., Schnorbus, M., Anslow, F., Sun, Q., Kirchmeier-Young, M., Zwiers, F., Seiler, C., Zhang, X., Flato, G., Wan, H., Li, G., Castellan, A., 2022. Human influence on the 2021 British Columbia floods. *Weather Clim. Extrem.* 36, 100441. <https://doi.org/10.1016/j.wace.2022.100441>.
- Gilroy, K.L., McCuen, R.H., 2012. A nonstationary flood frequency analysis method to adjust for future climate change and urbanization. *J. Hydrol.* 414–415, 40–48. <https://doi.org/10.1016/j.jhydrol.2011.10.009>.
- Giorgi, F., 2019. Thirty years of regional climate modeling: where are we and where are we going next? *J. Geophys. Res. Atmos.* 124, 5696–5723. <https://doi.org/10.1029/2018JD030094>.
- Giorgi, F., Bi, X., 2009. Time of emergence (TOE) of GHG-forced precipitation change hot-spots. *Geophys. Res. Lett.* 36. <https://doi.org/10.1029/2009GL037593>.
- Gizaw, M.S., Gan, T.Y., 2016. Regional flood frequency analysis using support vector regression under historical and future climate. *J. Hydrol.* 538, 387–398. <https://doi.org/10.1016/j.jhydrol.2016.04.041>.
- Griffis, V.W., Stedinger, J.R., 2007. The use of GLS regression in regional hydrologic analyses. *J. Hydrol.* 344, 82–95. <https://doi.org/10.1016/j.jhydrol.2007.06.023>.
- Gu, L., Chen, J., Xu, C.-Y., Kim, J.-S., Chen, H., Xia, J., Zhang, L., 2019. The contribution of internal climate variability to climate change impacts on droughts. *Sci. Total Environ.* 684, 229–246. <https://doi.org/10.1016/j.scitotenv.2019.05.345>.
- Guo, S., Xiong, L., Chen, J., Guo, S., Xia, J., Zeng, L., Xu, C.-Y., 2023. Nonstationary regional flood frequency analysis based on the Bayesian method. *Water Resour. Manag.* 37, 659–681. <https://doi.org/10.1007/s11269-022-03394-9>.
- Gupta, H.V., Kling, H., Yilmaz, K.K., Martinez, G.F., 2009. Decomposition of the mean squared error and NSE performance criteria: implications for improving hydrological modelling. *J. Hydrol.* 377, 80–91.
- Haddad, K., Rahman, A., 2012. Regional flood frequency analysis in eastern Australia: Bayesian GLS regression-based methods within fixed region and ROI framework—quantile regression vs. parameter regression technique. *J. Hydrol.* 430, 142–161. <https://doi.org/10.1016/j.jhydrol.2012.02.012>.
- Haddad, K., Rahman, A., 2011. Regional flood estimation in New South Wales Australia using generalized least squares quantile regression. *J. Hydrol. Eng.* 16, 920–925. [https://doi.org/10.1061/\(ASCE\)HE.1943-5584.0000395](https://doi.org/10.1061/(ASCE)HE.1943-5584.0000395).
- Haddad, K., Rahman, A., Kuczera, G., 2011. Comparison of ordinary and generalised least squares regression models in regional flood frequency analysis: a case study for New South Wales. *Australas. J. Water Resour.* 15, 59–70.
- Hailegeorgis, T.T., Alfredsen, K., 2017. Regional flood frequency analysis and prediction in ungauged basins including estimation of major uncertainties for mid-Norway. *J. Hydrol.: Reg. Stud.* 9, 104–126. <https://doi.org/10.1016/j.ejrh.2016.11.004>.
- Hamed, K., Rao, A.R., 2019. *Flood frequency analysis*. CRC Press.
- Han, X., Mehrotra, R., Sharma, A., Rahman, A., 2022. Incorporating nonstationarity in regional flood frequency analysis procedures to account for climate change impact. *J. Hydrol.* 612, 128235.
- Han, X., Ouara, T.B.M.J., Rahman, A., Haddad, K., Mehrotra, R., Sharma, A., 2020. A network approach for delineating homogeneous regions in regional flood frequency analysis. *Water Resour. Res.* 56, e2019WR025910. <https://doi.org/10.1029/2019WR025910>.
- Hastie, T., Friedman, J., Tibshirani, R., 2001. *The Elements of Statistical Learning*, Springer Series in Statistics. Springer New York, New York, NY. <https://doi.org/10.1007/978-0-387-21606-5>.
- Hawkins, E., Sutton, R., 2009. The potential to narrow uncertainty in regional climate predictions. *Bull. Am. Meteorol. Soc.* 90, 1095–1108.
- Hersbach, H., Bell, B., Berrisford, P., Hirahara, S., Horányi, A., Muñoz-Sabater, J., Nicolas, J., Peubey, C., Radu, R., Schepers, D., Simmons, A., Soci, C., Abdalla, S., Abellan, X., Balsamo, G., Bechtold, P., Bivatt, G., Bidlot, J., Bonavita, M., De Chiara, G., Dahlgren, P., Dee, D., Diamantakis, M., Dragani, R., Flemming, J., Forbes, R., Fuentes, M., Geer, A., Haimberger, L., Healy, S., Hogan, R.J., Hólm, E., Janisková, M., Keeley, S., Laloyaux, P., Lopez, P., Lupu, C., Radnoti, G., De Rosnay, P., Rozum, I., Vamborg, F., Villaume, S., Thépaut, J., 2020. The ERA5 global reanalysis. *Q. J. R. Meteorol. Soc.* 146, 1999–2049. <https://doi.org/10.1002/qj.3803>.
- Hirabayashi, Y., Mahendran, R., Koirala, S., Konoshima, L., Yamazaki, D., Watanabe, S., Kim, H., Kanae, S., 2013. Global flood risk under climate change. *Nat. Clim. Change* 3, 816–821. <https://doi.org/10.1038/nclimate1911>.
- Hoerl, A.E., Kennard, R.W., 1970. Ridge regression: applications to nonorthogonal problems. *Technometrics* 12, 69–82. <https://doi.org/10.1080/00401706.1970.10488635>.
- Hosking, J.R., 1990. L-moments: analysis and estimation of distributions using linear combinations of order statistics. *J. R. Stat. Soc. Ser. B Stat. Methodol.* 52, 105–124.
- Hosking, J.R.M., Wallis, J.R., 1997. Regional frequency analysis.
- Hosking, J.R.M., Wallis, J.R., 1993. Some statistics useful in regional frequency analysis. *Water Resour. Res.* 29, 271–281. <https://doi.org/10.1029/92WR01980>.
- Hu, L., Nikolopoulos, E.I., Marra, F., Anagnostou, E.N., 2020. Sensitivity of flood frequency analysis to data record, statistical model, and parameter estimation methods: an evaluation over the contiguous United States. *J. Flood Risk Manag.* 13, e12580. <https://doi.org/10.1111/jfr.12580>.
- Huo, W., Li, Z., Wang, J., Yao, C., Zhang, K., Huang, Y., 2019. Multiple hydrological models comparison and an improved Bayesian model averaging approach for ensemble prediction over semi-humid regions. *Stoch. Environ. Res. Risk Assess.* 33, 217–238.
- Innocenti, S., Mailhot, A., Leduc, M., Cannon, A.J., Frigon, A., 2019. Projected changes in the probability distributions, seasonality, and spatiotemporal scaling of daily and subdaily extreme precipitation simulated by a 50-member ensemble over Northeastern North America. *J. Geophys. Res. Atmos.* 124, 10427–10449. <https://doi.org/10.1029/2019JD031210>.
- Intergovernmental Panel on Climate Change (IPCC), 2021. *Climate Change 2021 – The Physical Science Basis: Working Group I Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press. <https://doi.org/10.1017/9781009157896>.

- Islam, M.R., Fereshtehpour, M., Najafi, M.R., Khaliq, M.N., Khan, A.A., Sushama, L., Khan, M.S., 2024. Climate-resilience of dams and levees in Canada: a review. *Discover Applied Sciences* 6 (4), 174. <https://doi.org/10.1007/s42452-024-05814-4>.
- Javelle, P., Ouara, T.B.M.J., Bobée, B., 2003. Spring flood analysis using the flood-duration-frequency approach: application to the provinces of Quebec and Ontario, Canada. *Hydrol. Process.* 17, 3717–3736. <https://doi.org/10.1002/hyp.1349>.
- Kay, J.E., Deser, C., Phillips, A., Mai, A., Hannay, C., Strand, G., Arblaster, J.M., Bates, S. C., Danabasoglu, G., Edwards, J., Holland, M., Kushner, P., Lamarque, J.-F., Lawrence, D., Lindsay, K., Middleton, A., Munoz, E., Neale, R., Oleson, K., Polvani, L., Vertenstein, M., 2015. The Community Earth System Model (CESM) large ensemble project: a community resource for studying climate change in the presence of internal climate variability. *Bull. Am. Meteorol. Soc.* 96, 1333–1349. <https://doi.org/10.1175/BAMS-D-13-00255.1>.
- Khaliq, M.N., Ouara, T.B., Ondo, J.-C., Gachon, P., Bobée, B., 2006. Frequency analysis of a sequence of dependent and/or non-stationary hydro-meteorological observations: a review. *J. Hydrol.* 329, 534–552. <https://doi.org/10.1016/j.jhydrol.2006.03.004>.
- Kidson, R., Richards, K.S., 2005. Flood frequency analysis: assumptions and alternatives. *Prog. Phys. Geogr.* 29, 392–410.
- Kim, H., Shin, J.-Y., Kim, T., Kim, S., Heo, J.-H., 2020. Regional frequency analysis of extreme precipitation based on a nonstationary population index flood method. *Adv. Water Resour.* 146, 103757. <https://doi.org/10.1016/j.advwatres.2020.103757>.
- King, A.D., Donat, M.G., Fischer, E.M., Hawkins, E., Alexander, L.V., Karoly, D.J., Dittus, A.J., Lewis, S.C., Perkins, S.E., 2015. The timing of anthropogenic emergence in simulated climate extremes. *Environ. Res. Lett.* 10, 094015. <https://doi.org/10.1088/1748-9326/10/9/094015>.
- Koutsoyiannis, D., 2023. Stochastics of hydroclimatic extremes - a cool look at risk. *Kallipos*. <https://doi.org/10.57713/kallipos-1>.
- Koutsoyiannis, D., Montanari, A., 2022. Climate extrapolations in hydrology: the expanded BlueCat methodology. *Hydrology* 9, 86.
- Kutner, M.H., Nachtsheim, C.J., Neter, J., Li, W., 2005. *Applied linear statistical models*. McGraw-hill.
- Kwon, Y.-O., Deser, C., 2007. North pacific decadal variability in the community climate system model version 2. *J. Clim.* 20, 2416–2433. <https://doi.org/10.1175/JCLI4103.1>.
- Leclerc, M., Ouara, T.B.M.J., 2007. Non-stationary regional flood frequency analysis at ungauged sites. *J. Hydrol.* 343, 254–265. <https://doi.org/10.1016/j.jhydrol.2007.06.021>.
- Lehner, F., Deser, C., Terray, L., 2017. Toward a new estimate of “time of emergence” of anthropogenic warming: insights from dynamical adjustment and a large initial-condition model ensemble. *J. Clim.* 30, 7739–7756. <https://doi.org/10.1175/JCLI-D-16-0792.1>.
- Li, M., Zhang, T., Feng, P., 2019. A nonstationary runoff frequency analysis for future climate change and its uncertainties. *Hydrol. Process.* 33, 2759–2771. <https://doi.org/10.1002/hyp.13526>.
- Lima, C.H.R., Lall, U., Troy, T., Devineni, N., 2016. A hierarchical Bayesian GEV model for improving local and regional flood quantile estimates. *J. Hydrol.* 541, 816–823. <https://doi.org/10.1016/j.jhydrol.2016.07.042>.
- Mangukiyi, N.K., Sharma, A., 2024. Alternate pathway for regional flood frequency analysis in data-sparse region. *J. Hydrol.* 629, 130635. <https://doi.org/10.1016/j.jhydrol.2024.130635>.
- Mann, H.B., 1945. Nonparametric tests against trend. *Econom. J. Econom. Soc.* 245–259.
- Maraun, D., 2016. Bias correcting climate change simulations - a critical review. *Curr. Clim. Change Rep.* 2, 211–220. <https://doi.org/10.1007/s40641-016-0050-x>.
- Menéndez, M., Woodworth, P.L., 2010. Changes in extreme high water levels based on a quasi-global tide-gauge data set. *J. Geophys. Res. Oceans* 115. <https://doi.org/10.1029/2009JC005997>.
- Merz, R., Blöschl, G., 2009. Process controls on the statistical flood moments - a data based analysis. *Hydrol. Process.* 23, 675–696. <https://doi.org/10.1002/hyp.7168>.
- Merz, R., Blöschl, G., 2005. Flood frequency regionalisation—spatial proximity vs. catchment attributes. *J. Hydrol.* 302, 283–306.
- Mosaffaie, J., 2015. Comparison of two methods of regional flood frequency analysis by using L-moments. *Water Resour.* 42, 313–321. <https://doi.org/10.1134/S0097807815030112>.
- Msilini, A., Charron, C., Ouara, T.B.M.J., Masselot, P., 2022. Flood frequency analysis at ungauged catchments with the GAM and MARS approaches in the Montreal region, Canada. *Can. Water Resour. J. Rev. Can. Ressour. Hydr.* 47, 111–121. <https://doi.org/10.1080/07011784.2022.2044385>.
- Musselman, K.N., Lehner, F., Ikeda, K., Clark, M.P., Prein, A.F., Liu, C., Barlage, M., Rasmussen, R., 2018. Projected increases and shifts in rain-on-snow flood risk over western North America. *Nat. Clim. Change* 8, 808–812. <https://doi.org/10.1038/s41558-018-0236-4>.
- Najafi, M.R., Moradkhani, H., 2014. A hierarchical Bayesian approach for the analysis of climate change impact on runoff extremes. *Hydrol. Process.* 28, 6292–6308. <https://doi.org/10.1002/hyp.10113>.
- Najafi, M.R., Moradkhani, H., 2013. Analysis of runoff extremes using spatial hierarchical Bayesian modeling. *Water Resour. Res.* 49, 6656–6670. <https://doi.org/10.1002/wrcr.20381>.
- Najafi, M.R., Moradkhani, H., Jung, I.W., 2011. Assessing the uncertainties of hydrologic model selection in climate change impact studies. *Hydrol. Process.* 25, 2814–2826. <https://doi.org/10.1002/hyp.8043>.
- Nash, J.E., Sutcliffe, J.V., 1970. River flow forecasting through conceptual models part I—a discussion of principles. *J. Hydrol.* 10, 282–290.
- Nguyen, T.-H., Min, S.-K., Paik, S., Lee, D., 2018. Time of emergence in regional precipitation changes: an updated assessment using the CMIP5 multi-model ensemble. *Clim. Dyn.* 51, 3179–3193. <https://doi.org/10.1007/s00382-018-4073-y>.
- O'Brien, N.L., Burn, D.H., 2014. A nonstationary index-flood technique for estimating extreme quantiles for annual maximum streamflow. *J. Hydrol.* 519, 2040–2048. <https://doi.org/10.1016/j.jhydrol.2014.09.041>.
- Ontario watershed information tool (OWIT), 2022.
- Ouali, D., Chebana, F., Ouara, T.B., 2016. Quantile regression in regional frequency analysis: a better exploitation of the available information. *J. Hydrometeorol.* 17, 1869–1883.
- Ouara, T.B.M.J., Bâ, K.M., Diaz-Delgado, C., Cârsteanu, A., Chokmani, K., Gingras, H., Quentin, E., Trujillo, E., Bobée, B., 2008. Intercomparison of regional flood frequency estimation methods at ungauged sites for a Mexican case study. *J. Hydrol.* 348, 40–58. <https://doi.org/10.1016/j.jhydrol.2007.09.031>.
- Ouara, T.B.M.J., Charron, C., 2019. Changes in the distribution of hydro-climatic extremes in a non-stationary framework. *Sci. Rep.* 9, 8104. <https://doi.org/10.1038/s41598-019-44603-7>.
- Pall, P., Aina, T., Stone, D.A., Stott, P.A., Nozawa, T., Hilberts, A.G., Lohmann, D., Allen, M.R., 2011. Anthropogenic greenhouse gas contribution to flood risk in England and Wales in autumn 2000. *Nature* 470, 382–385.
- Pandey, G.R., Nguyen, V.-T.-V., 1999. A comparative study of regression based methods in regional flood frequency analysis. *J. Hydrol.* 225, 92–101. [https://doi.org/10.1016/S0022-1694\(99\)00135-3](https://doi.org/10.1016/S0022-1694(99)00135-3).
- Peel, M.C., Wang, Q.J., Vogel, R.M., McMahon, T.A., 2001. The utility of L-moment ratio diagrams for selecting a regional probability distribution. *Hydrol. Sci. J.* 46, 147–155. <https://doi.org/10.1080/02626660109492806>.
- Poitras, V., Sushama, L., Seglenieks, F., Khaliq, M.N., Soulis, E., 2011. Projected changes to streamflow characteristics over western Canada as simulated by the Canadian RCM. *J. Hydrometeorol.* 12, 1395–1413. <https://doi.org/10.1175/JHM-D-10-05002.1>.
- Raftery, A.E., Gneiting, T., Balabdaoui, F., Polakowski, M., 2005. Using Bayesian model averaging to calibrate forecast ensembles. *Mon. Weather Rev.* 133, 1155–1174. <https://doi.org/10.1175/MWR2906.1>.
- RahimiMovaghar, M., Najafi, M.R., Shrestha, R.R., Liu, Y., 2025. Successive warm-wet and warm-dry events in the Great Lakes Basin: future projections using CMIP6 models. *Clim. Dyn.* 63, 80. <https://doi.org/10.1007/s00382-024-07565-9>.
- Rahman, A.S., Khan, Z., Rahman, A., 2020. Application of independent component analysis in regional flood frequency analysis: comparison between quantile regression and parameter regression techniques. *J. Hydrol.* 581, 124372. <https://doi.org/10.1016/j.jhydrol.2019.124372>.
- Reinders, J.B., Munoz, S.E., 2024. Accounting for hydroclimatic properties in flood frequency analysis procedures. *Hydrol. Earth Syst. Sci.* 28, 217–227.
- Salas, J.D., Obeysekere, J., 2014. Revisiting the concepts of return period and risk for nonstationary hydrologic extreme events. *J. Hydrol. Eng.* 19, 554–568. [https://doi.org/10.1061/\(ASCE\)HE.1943-5584.0000820](https://doi.org/10.1061/(ASCE)HE.1943-5584.0000820).
- Sampaio, J., Costa, V., 2021. Bayesian regional flood frequency analysis with GEV hierarchical models under spatial dependency structures. *Hydrol. Sci. J.* 66, 422–433. <https://doi.org/10.1080/02626667.2021.1873997>.
- Satterthwaite, D., 2009. The implications of population growth and urbanization for climate change. *Environ. Urban.* 21, 545–567.
- Schmeits, M.J., Kok, K.J., 2010. A comparison between raw ensemble output, (modified) Bayesian model averaging, and extended logistic regression using ECMWF ensemble precipitation reforecasts. *Mon. Weather Rev.* 138, 4199–4211.
- Schneider, E.K., Kinter, J.L., 1994. An examination of internally generated variability in long climate simulations. *Clim. Dyn.* 10, 181–204. <https://doi.org/10.1007/BF00208987>.
- Scinocca, J.F., Khari, V.V., Jiao, Y., Qian, M.W., Lazare, M., Solheim, L., Flato, G.M., Biner, S., Desgagne, M., Dugas, B., 2016. Coordinated global and regional climate modeling. *J. Clim.* 29, 17–35. <https://doi.org/10.1175/JCLI-D-15-0161.1>.
- Serinaldi, F., Kilsby, C.G., 2018. Unsurprising surprises: the frequency of record-breaking and overthreshold hydrological extremes under spatial and temporal dependence. *Water Resour. Res.* 54, 6460–6487. <https://doi.org/10.1029/2018WR023055>.
- Serinaldi, F., Kilsby, C.G., Lombardo, F., 2018. Untenable nonstationarity: an assessment of the fitness for purpose of trend tests in hydrology. *Adv. Water Resour.* 111, 132–155.
- Shapely, L.S., 1953. A value for n-person games. *Contrib. Theory Games II Ann. Math. Stud.* 28, 307–317.
- Sharifi Garmdareh, E., Vafakhah, M., Eslamian, S.S., 2018. Regional flood frequency analysis using support vector regression in arid and semi-arid regions of Iran. *Hydrol. Sci. J.* 63, 426–440.
- Singh, N.K., Emanuel, R.E., McGlynn, B.L., Miniati, C.F., 2021. Soil moisture responses to rainfall: Implications for runoff generation. *Water Resources Research* 57 (9), e2020WR028827. <https://doi.org/10.1029/2020WR028827>.
- Singh, J., Ghosh, S., Simonovic, S.P., Karmakar, S., 2021. Identification of flood seasonality and drivers across Canada. *Hydrological Processes* 35 (10), e14398. <https://doi.org/10.1002/hyp.14398>.
- Singh, H., Najafi, M.R., Cannon, A.J., 2021. Characterizing non-stationary compound extreme events in a changing climate based on large-ensemble climate simulations. *Climate Dynamics* 56 (5), 1389–1405. <https://doi.org/10.1007/s00382-020-05538-2>.
- Sloughter, J.M.L., Raftery, A.E., Gneiting, T., Fraley, C., 2007. Probabilistic quantitative precipitation forecasting using Bayesian model averaging. *Mon. Weather Rev.* 135, 3209–3220.
- Stocker, T.F., Qin, D., Plattner, G.-K., Alexander, L.V., Allen, S.K., Bindoff, N.L., Bréon, F.-M., Church, J.A., Cubasch, U., Emori, S., 2013. Technical summary, in: *Climate Change 2013: The Physical Science Basis*. Contribution of Working Group I to the

- Fifth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge University Press, pp. 33–115.
- Sui, Y., Lang, X., Jiang, D., 2014. Time of emergence of climate signals over China under the RCP4.5 scenario. *Clim. Change* 125, 265–276. <https://doi.org/10.1007/s10584-014-1151-y>.
- Sushanth, K., Mishra, A., Mukhopadhyay, P., Singh, R., 2023. Real-time streamflow forecasting in a reservoir-regulated river basin using explainable machine learning and conceptual reservoir module. *Sci. Total Environ.* 861, 160680.
- Szeto, K., Brimelow, J., Gysbers, P., Stewart, R., 2015. 5. The 2014 extreme flood on the southeastern Canadian prairies. *Bull. Am. Meteorol. Soc.* 96, S20–S24.
- Teufel, B., Diro, G.T., Whan, K., Milrad, S.M., Jeong, D.I., Ganji, A., Huziy, O., Winger, K., Gyakum, J.R., De Elia, R., 2017. Investigation of the 2013 Alberta flood from weather and climate perspectives. *Clim. Dyn.* 48, 2881–2899. <https://doi.org/10.1007/s00382-016-3239-8>.
- Thomas, D.M., Benson, M.A., 1970. Generalization of streamflow characteristics from drainage-basin characteristics. US Government Printing Office, Washington, DC, USA.
- Tibshirani, R., 1996. Regression shrinkage and selection via the lasso. *J. R. Stat. Soc. Ser. B Stat. Methodol.* 58, 267–288.
- Villarini, G., Smith, J.A., Serinaldi, F., Bales, J., Bates, P.D., Krajewski, W.F., 2009. Flood frequency analysis for nonstationary annual peak records in an urban drainage basin. *Adv. Water Resour.* 32, 1255–1266. <https://doi.org/10.1016/j.advwatres.2009.05.003>.
- Vincent, L.A., Zhang, X., Brown, R.D., Feng, Y., Mekis, E., Milewska, E.J., Wan, H., Wang, X.L., 2015. Observed trends in Canada's climate and influence of low-frequency variability modes. *J. Clim.* 28, 4545–4560.
- Vu, T.M., Mishra, A.K., 2019. Nonstationary frequency analysis of the recent extreme precipitation events in the United States. *J. Hydrol.* 575, 999–1010. <https://doi.org/10.1016/j.jhydrol.2019.05.090>.
- Warden, J.W., Rezvani, R., Najafi, M.R., Shrestha, R.R., 2024. Projections of rain-on-snow events in a sub-arctic river basin under 1.5° C–4° C global warming. *Hydrological Processes* 38 (8), e15250. <https://doi.org/10.1002/hyp.15250>.
- Westra, S., Fowler, H.J., Evans, J.P., Alexander, L.V., Berg, P., Johnson, F., Kendon, E.J., Lenderink, G., Roberts, N.M., 2014. Future changes to the intensity and frequency of short-duration extreme rainfall. *Rev. Geophys.* 52, 522–555. <https://doi.org/10.1002/2014RG000464>.
- Yan, H., Moradkhani, H., 2015. A regional Bayesian hierarchical model for flood frequency analysis. *Stoch. Environ. Res. Risk Assess.* 29, 1019–1036. <https://doi.org/10.1007/s00477-014-0975-3>.
- Yin, J., Gentile, P., Zhou, S., Sullivan, S.C., Wang, R., Zhang, Y., Guo, S., 2018. Large increase in global storm runoff extremes driven by climate and anthropogenic changes. *Nat. Commun.* 9, 4389. <https://doi.org/10.1038/s41467-018-06765-2>.
- Zahmatkesh, Z., Kumar Jha, S., Coulibaly, P., Stadnyk, T., 2019. An overview of river flood forecasting procedures in Canadian watersheds. *Can. Water Resour. J. Rev. Can. Ressour. Hydr.* 44, 213–229. <https://doi.org/10.1080/07011784.2019.1601598>.
- Zhang, Z., Stadnyk, T.A., Burn, D.H., 2020. Identification of a preferred statistical distribution for at-site flood frequency analysis in Canada. *Can. Water Resour. J. Rev. Can. Ressour. Hydr.* 45, 43–58. <https://doi.org/10.1080/07011784.2019.1691942>.